

The Paired-Delta Protocol: A Within-Call Reference Architecture for Observer–QRNG Research

By Andrea Rester Campbell

Independent Researcher

Port Angeles, WA USA

Abstract

Experiments testing whether human intention influences quantum random number generators (QRNGs) face persistent credibility and measurement problems. Small reported effects are difficult to separate from publication bias, analytic flexibility, hardware variation, timing effects, and environmental fluctuation. We introduce the Paired-Delta Protocol (PDP), which derives a Subject stream and Paired Control Stream (PCS) from the same QRNG call. PDP is metric-agnostic and supports three complementary views: (1) conventional direct-stream analysis, (2) paired-differential analysis of Subject minus PCS, and (3) relational analysis of their association, temporal alignment, and stability. The paired-differential and relational views have no single-channel equivalent.

The pilot comprised 11,607 blocks across Human, AI, and automated Baseline conditions, including 121 Human participants. Baseline diagnostics detected no same-call association at the resolutions tested. In controlled injections, specified

equal-strength disturbances produced little systematic displacement of the paired difference, while stream-specific and unequal disturbances remained detectable. No naturally occurring disturbance of known type and magnitude was observed, so operational artifact cancellation was not established.

No clear condition differences were detected in paired mean hit rate. An exploratory sequence-ordering difference appeared in a retrospectively selected subgroup of three participants. Conventional analysis showed a small upward Subject-stream difference from Baseline, which could have been interpreted as a candidate observer-linked result. PDP showed that the larger paired contrast was formed about equally by Subject above and PCS below their respective Baseline values and was enlarged by the particular paired Baseline observed. The direct and paired findings therefore did not provide independent support for that interpretation, and the opposed-stream pattern was compatible with selection for a large paired difference. These distinctions prevented a more favorable but incomplete interpretation based on the Subject stream alone.

The primary contribution is increased diagnostic leverage. PDP makes otherwise inaccessible two-stream phenomena observable and provides additional checks on conventional single-stream findings. It reveals how Subject, PCS, and the empirical Baseline contribute to a contrast and enables analysis of the streams' association, temporal alignment, and stability. These capabilities can uncover new patterns and help prevent ambiguous findings from appearing stronger than the evidence supports, thereby addressing a central problem in observer-QRNG research.

1. Introduction

1.1 The Credibility Problem in Observer–QRNG Research

Decades of experiments testing whether human intention influences quantum or electronic random number generators have produced a large but contested body of data (e.g., Jahn et al., 1997). The central scientific disputes in this literature are primarily about the credibility of the aggregate evidence. The most influential skeptical analysis, Bösch, Steinkamp, and Boller's (2006) meta-analysis of 380 studies in *Psychological Bulletin*, found a statistically significant but extremely small overall effect that was strongly and inversely related to sample size and highly heterogeneous. They argued it is a pattern consistent with publication bias and selective reporting rather than a genuine effect. The ensuing exchange (Radin, Nelson, Dobyns, & Houtkooper, 2006; Wilson & Shadish, 2006) disputed the interpretation but not the underlying concern: that small effects in this paradigm are difficult to separate from the combined influence of analytic flexibility, selective reporting, and publication bias. More recent pre-registered work, adequately powered to detect an effect if one existed, reported Bayesian evidence against micro-psychokinetic effects on QRNG output (Maier, Dechamps, & Pflitsch, 2018), reinforcing that the field's central unresolved question is whether reported effects are credible and reproducible and not whether current methods are capable of detecting a real one.

A second, separate difficulty is that even a well-powered, single-channel design cannot show how an observed deviation is distributed across parts of the same

acquisition. It cannot determine whether the deviation is confined to the retained stream, shared across the acquisition, or accompanied by changes in a same-call companion stream. In distributed RNG networks such as the Global Consciousness Project, the project's own researchers have identified electromagnetic interference and other environmental factors as plausible alternative explanations for observed deviations (Nelson & Bancel, 2011). PDP addresses this ambiguity by adding a within-call reference. In controlled injection tests, contributions applied equally to both stream statistics produced little systematic displacement of the paired difference, while stream-specific and unequal contributions remained detectable. Future applications can evaluate the same operating principle against independently characterized natural artifacts.

This points to two distinct problems that any single study in this paradigm must confront.

1.1.1 Problem 1: Cross-study credibility

The dominant critiques concern the research process. Undisclosed analytic choices, outcome-dependent stopping, and selective publication together can manufacture small apparent effects across a literature. The accepted remedies are procedural, including pre-registration of hypotheses and analyses, locked datasets, and transparent reporting, all applied before outcomes are known.

1.1.2 Problem 2: Within-study artifact ambiguity

Separately, within any individual study, expected deviations may be small relative to variation in hardware behavior, timing, and environmental conditions. Increasing sample size does not resolve this ambiguity because it raises sensitivity to genuine signals and systematic confounds alike. A single retained stream also cannot show whether a deviation is shared across or differs between parts of the same acquisition. A prespecified within-call comparison adds diagnostic leverage by removing contributions expressed equally in both selected stream statistics and retaining contributions that differ between them. It does not identify the cause of a surviving difference, but it provides component and relational information unavailable from a single-channel design.

This paper addresses both problems, though unevenly. For Problem 1 (credibility) the pilot's safeguards include: metrics were specified in the experiment code before data collection, and the dataset was frozen with cryptographic fingerprints prior to analysis. All findings are treated as exploratory. The complete remedy for this problem is prospective pre-registration, Exp 5, which this pilot motivates but does not itself provide (Section 4.4).

For Problem 2, we introduce a Paired-Delta Protocol that builds a within-call reference into every observation (Section 1.2). This reference is a prespecified feature of data generation rather than a control constructed or matched after outcomes are known. The architecture provides two forms of added value. It adds diagnostic checks to conventional stream-level findings and makes otherwise

inaccessible two-stream relationships measurable. These capabilities are introduced in Section 1.5 and demonstrated in Sections 3 and 4.

1.2 The Paired-Delta Architecture

The Paired-Delta Protocol is a data-generation and analysis architecture for experiments in which small differences may be difficult to distinguish from ordinary variation in the measurement system. Each acquisition event produces two streams from the same underlying draw. One stream is designated the Subject stream and the other the Paired Control Stream, or PCS. The first bit of each call determines which physical stream receives each label. This assignment rule is fixed before analysis.

The architecture is not defined by one particular stream length, partition rule, QRNG provider, or metric. A study may divide or transform each acquisition in different ways, provided that the rule is specified in advance and its effects are calibrated. The choice of stream construction is part of the study design.

The same-call data support three forms of analysis. Each stream can be analyzed separately. The two stream-level values can be compared within the same acquisition. The two streams can also be analyzed jointly to study their relationship. These analyses use the same within-call data but answer different questions.

The architecture distinguishes two collected reference types. There is the Paired Control Stream (PCS) which is generated within each call, and an empirical

between-condition Baseline. All metrics offer an additional comparison to a theoretical chance value, although some metrics may require calibration to determine the reference point.

The within-call comparison can remove a contribution when it changes the two stream statistics equally. Contributions that affect the streams unequally, vary across the acquisition, or interact with the selected metric may remain. The same limitation applies to any genuine influence that affects both stream statistics equally. The paired comparison therefore provides conditional common-mode control rather than general protection from all artifacts.

A paired comparison also has a precision cost. When the two stream statistics are independent and have similar variability, their difference is more variable than either stream statistic alone. The direct stream values remain available for examining changes that the paired comparison could remove.

$$\Delta_M = M_{Subject} - M_{PCS}.$$

1.3 Experiment 4 as a Pilot Implementation

Experiment 4 was a pilot implementation of the Paired-Delta architecture. Its settings were chosen to make the study feasible with limited access to paid participants, API calls, and longer acquisitions. The pilot used 301-bit QRNG calls. Bit 0 served as the assignment bit. The remaining 300 bits were divided into two

contiguous 150-bit physical streams. The assignment bit determined which stream received the Subject label and which received the PCS label.

Participants saw feedback based only on the Subject stream. The PCS was not displayed and served as the within-call reference. After every fifth experimental block, a separate 1,000-bit QRNG audit call was made without a target display, intention, or feedback. These calls provided an independent check on QRNG behavior across the study. They did not pass through the full experimental splitting and scoring pathway.

The pilot also included an AI condition as a pilot-specific procedural comparison. The AI condition was used to examine whether the task interface, timing structure, and intent-framed prompting produced patterns distinguishable from the Human and Baseline conditions. It is not a required component of the Paired-Delta Protocol.

These settings describe the pilot implementation and are not requirements of the general protocol. A future study may use a different stream length, partition rule, acquisition design, or metric. Such choices must be specified before data collection begins, since the protocol requires calibration for the selected stream length, partition rule, acquisition design, and metric.

Figure 1. Paired-Delta QRNG architecture in Pilot

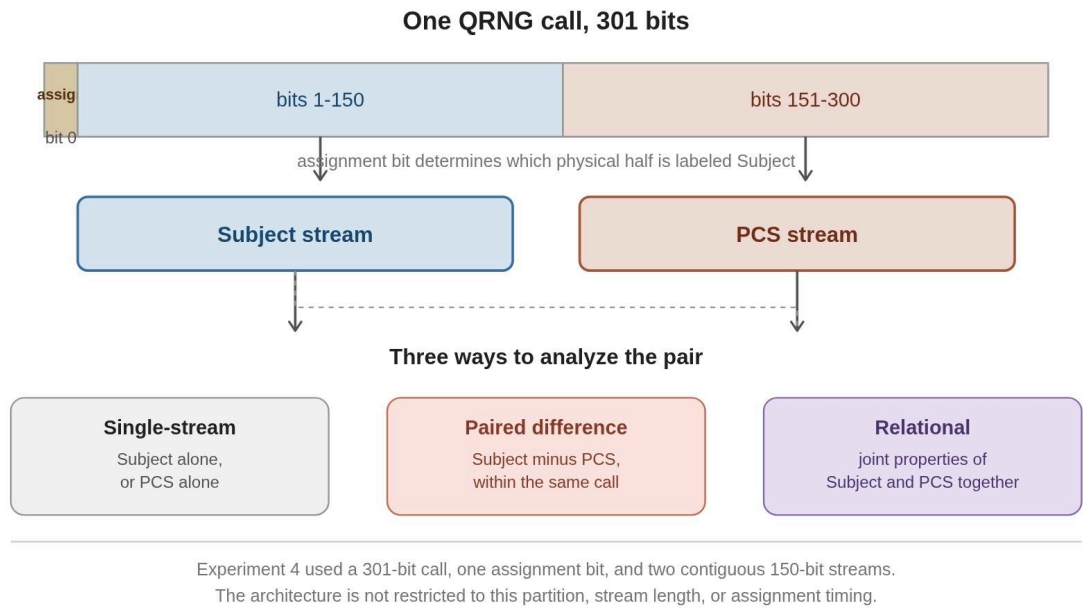


Figure 1. In the pilot a single QRNG call generates an ordered sequence that is divided into two physical streams. A random assignment rule labels one stream Subject and the other PCS. It used a 301-bit call with one assignment bit and two contiguous 150-bit streams. The architecture supports separate stream-level analysis, within-call paired differences, and joint analysis of the two streams.

Both hit rate and the single-scale H_{RS} score were available in the original experiment code. H_{RS} became the main worked example because it could examine ordering within the short streams and produced a pattern that warranted closer examination (Section 1.4).

The pilot used an empirical Baseline condition that consisted of automated blocks run on the same system without a Subject present, as the between-condition comparison for direct, paired, and relational analyses.

1.4 Exploratory and Confirmatory Use

PDP can be used both when a researcher has a defined hypothesis and when the researcher is still exploring which features or measures are informative.

In a confirmatory study, the researcher specifies the proposed effect, the stream construction, the metric, the analysis family, and the comparison before examining the outcome data. The study also specifies the calibration procedure, grouping structure, direction of the test, and decision rule. This use is appropriate when previous work or theory identifies a particular pattern for testing.

In an exploratory study, the researcher may not yet know which metric, stream property, or relationship will be most informative. PDP can support a structured capability sweep across direct, paired, and relational analyses. The purpose is to examine the available analytical space, identify useful measures, reveal redundant measures, detect finite-sample limitations, and identify candidate patterns for future study. Its value is that it makes the search visible and gives the next study a more focused design. Candidate measures and patterns selected from the sweep should be evaluated in a new study with a prespecified analysis plan.

The pilot used this exploratory mode. The pilot was designed primarily to test the methodology across a broad set of measures, with no strong expectation that any

candidate pattern would emerge. It evaluated the implementation, tested the behavior of paired subtraction under specified simulated disturbances, examined a broad set of direct, paired, and relational measures, and identified calibration and interpretation problems that matter for future work.

The H_{RS} pattern is therefore presented as a worked example of the exploratory process. It illustrates how a candidate pattern can emerge from the analysis sweep, be examined through complementary analyses, and lead to a more focused prospective study. It is not presented as a confirmed Human-specific result.

1.5 The Three Analysis Families

PDP supports three analysis families.

Single-stream analysis examines one stream at a time. A researcher may calculate a measure for the Subject stream or for PCS and compare that measure across conditions. This can reveal a condition-level change in one stream. It does not by itself show whether the change is specific to the Subject stream or shared by both streams.

Paired-stream difference analysis compares the two streams within each acquisition. For a selected stream-level measure M , the paired difference is

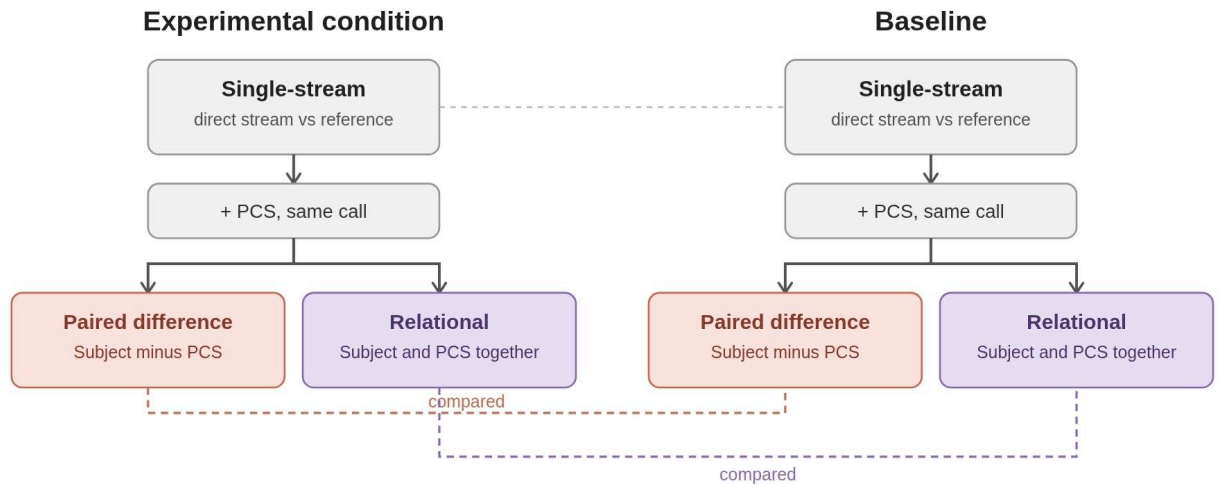
$$\Delta_M = M_{Subject} - M_{PCS}.$$

This asks whether the labeled Subject stream differs from its same-call PCS reference. It can reduce contributions expressed equally in both stream statistics. It can also remove a whole-call change that affects both streams equally.

Relational analysis examines the two streams jointly. It can ask whether the streams are associated, whether their association varies across sessions or local windows, or whether their relationship changes across temporal offsets. The central question is comparative and asks whether this joint behavior differs when a Subject is present relative to the automated Baseline condition. A relational measure asks how the streams vary together rather than whether the two stream values differ in level.

Each analysis family can be applied within the experimental and Baseline conditions and then compared across conditions. Each analysis family can be applied within the experimental and Baseline conditions and then compared across conditions. Because the analyses use overlapping data from the same calls, they should not be treated as statistically independent confirmations. Their joint interpretation is nevertheless informative because each answers a different question. Direct and paired-differential analyses show how the stream-level components contribute to an observed contrast, while relational analysis examines properties of the pair that are not defined for either stream alone.

Figure 2. PDP Analysis taxonomy



Each analysis family is computed identically for the experimental condition and for Baseline, then compared using its corresponding quantity. Single-stream analysis is available to any single-channel design; paired difference and relational analysis require two streams from the same call.

Figure 2. The same-call data support single-stream, paired-stream difference, and relational analyses. Each analysis family can be compared across experimental conditions using its corresponding quantity. PCS is the within-call reference for the paired difference. Baseline is the between-condition empirical reference.

2. Methods

2.1 Participants

A total of 121 human participants contributed data. Recruitment occurred through two main channels plus a small amount of personal outreach.

Most participants (approximately 100) were recruited through a survey-exchange platform where users complete other researchers' studies in exchange for credits to post their own. This group participated primarily for platform incentives and had no special reason to care about the topic beyond completing the task.

A smaller group was recruited through informal posts to Facebook and to the Reddit community r/remoterviewing. Facebook recruitment produced only a handful of participants, while the Reddit community provided most of the non-survey-exchange participants. In the Reddit outreach, participants were explicitly encouraged to complete five to ten sessions, because repeated participation was expected to be informative in a short-form protocol.

No exclusion criteria were applied prior to data collection.

Engagement was self-selected. Most participants completed a single session, a small number completed multiple sessions, and three participants completed five or more sessions. One of the three ≥ 5 -session participants was the principal investigator, and contributed the most sessions (18) in that subgroup. Analyses that rely on the ≥ 5 -session subgroup are treated as exploratory and are accompanied by descriptive checks of whole-session recurrence, leave-one-whole-session-out influence, and a numerical comparison of the pooled estimate with the estimate obtained after excluding the principal investigator. These checks assess recurrence and concentration within the observed subgroup but do not establish generalizability across participants.

Because recruitment source and engagement level are confounded in this pilot dataset, session-count effects cannot be interpreted as learning or “dose response.” Higher-session participants were disproportionately drawn from the Reddit recruitment, while single-session participants were disproportionately drawn from the survey-exchange platform. Separating repeated exposure from population self-selection requires a replication design that recruits both sources with balanced repeat-participation targets.

Intake data collected prior to each session included age and gender; ADHD and autism-spectrum diagnoses; a set of attention-related self-reports (daily screen time, frequency of attentional lapses, time in deep-focus activities, and frequency of flow states); and psi- and state-related items (meditation frequency, openness to psi possibility, and current mental clarity). Of these, meditation frequency and openness to psi possibility (the sheep-goat variable, contrasting psi believers with skeptics) were analyzed as exploratory moderators of hit rate and ΔH_{RS} (ordering-structure) metrics. The remaining items were collected for sample characterization and are reserved for analysis in Exp 5, where the larger sample provides adequate power. These variables are included in the publicly available dataset for future analysis. Post-session ratings of focus, calm, and confidence, along with optional free-text feedback, were recorded but not analyzed.

2.2 Apparatus

Quantum random bits were generated using hardware quantum random number generators (QRNGs) accessed via public web APIs. Due to API rate limits and service availability constraints, three independent QRNG services were used sequentially during data collection: Outshift (Cisco Quantum Entropy Service), LFDR QRNG (Laboratory for Digitalisation Research; Germany), and the ANU Quantum Random Numbers Server (Australian National University). The majority of blocks were collected using the first two providers.

Each experimental block fetched 301 bits in a single API call. Both the Subject stream and the Paired Control Stream (PCS) were derived from sequential halves of the same 301-bit call. This same-call origin allowed paired subtraction to remove contributions expressed equally in the two calculated stream statistics, but did not ensure that device characteristics affected the two halves or statistics equally.

QRNG provider identity was not stored at the block level because a code error captured a reference rather than its value, and it therefore could not be included as a block-level analytic factor. Provider provenance could be partially reconstructed from session and audit records, but it was incomplete and partly confounded with collection period and condition. Same-call pairing may reduce provider-related contributions expressed equally in both stream statistics, but it does not automatically eliminate provider effects.

The experiment ran in a web browser. For human participants, a target color was displayed and held constant across each 5-block interval, re-randomized at each audit call. Once the participant initiated the QRNG call, the target pulsed at 5 Hz during bit generation.

2.3 Procedure

Each session was designed to contain 30 blocks. A target color was assigned at the start of each 5-block interval and held constant across those blocks, with re-randomization occurring at each audit call (every fifth block). When ready, the participant pressed a button to initiate a single 301-bit QRNG call. During bit generation, the target pulsed at 5 Hz while the participant maintained focus.

Bit 0 served as the assignment bit, randomly determining which 150-bit half of the call was labeled the Subject stream and which was labeled the Paired Control Stream (PCS). Participants never saw the raw bitstream. After completion of the call, feedback was displayed as the hit rate of the Subject stream (proportion of bits matching the intended direction). The PCS was scored identically but not displayed.

Every fifth block, a separate 1000-bit audit call was made without task context (no target, no intention, no feedback). These audit calls provided separate diagnostic measurements of QRNG behavior during data collection. Because they did not use the 301-bit trial-call processing path, they were analyzed separately from the experimental blocks.

2.4 Conditions

The pilot included Human, AI, and Baseline conditions. The Human and Baseline conditions provided the main experimental comparison. The AI condition was a pilot-specific procedural comparison and is not a required component of the Paired-Delta Protocol.

2.4.1 Human

Participants performed the intention task described in Section 2.3. They initiated each block, focused on the displayed target, and received feedback based on the Subject stream.

2.4.2 AI Agent

The experiment included an AI-driven condition implemented using GPT-4o-mini. This condition served two purposes. First, it provided a procedural control for task flow, timing, and interface dynamics under the paired-delta architecture. Second, it allowed an exploratory assessment of whether a non-biological agent, when given intent-framed prompts and feedback, would exhibit any measurable divergence from hardware baseline or any learning-like dynamics across blocks or sessions.

The AI condition consisted of two components: GPT-4o-mini, which generated text responses to prompts, and a Puppeteer automation layer that executed the corresponding browser actions. On each block, Puppeteer detected the “Ready” screen and provided GPT with a prompt mirroring what a human participant would

see (e.g., “Focus intention on BLUE. Ready.”). After GPT returned a response, Puppeteer initiated the QRNG call. During the “Fetching quantum data...” phase, GPT received a second prompt aligned with the pulsing target display. After each block, the agent received a reduced form of the feedback available to human participants (its hit-rate percentage for that block) before proceeding.

Despite this alignment, the AI condition differs from human participation in important ways. Human participants are self-paced and may delay initiation, adjust their internal state, or pause between blocks. GPT, by contrast, operates within the lifecycle of an API call and cannot intentionally delay or modulate timing beyond the duration of its response generation. Although the agent’s output gates the initiation of each QRNG call, it does not exercise open-ended control over when initiation occurs. This implementation represents the closest analogue achievable within a stateless API-driven framework, while acknowledging that it is not equivalent to human self-pacing.

In the present study, the AI condition therefore functions both as a procedural control and as an exploratory probe of whether intent-framed prompting in a non-biological system produces any detectable structure under the paired-delta architecture.

2.4.3 Baseline

The Baseline condition used fully automated execution without an agent present to receive or act on the intention framing. Block initiation and inter-block timing were scripted rather than self-paced.

Baseline provided the between-condition empirical reference for the direct, paired, and relational analyses. It was not an ideal theoretical reference and was not collected contemporaneously with the Human sessions. The implications of the different collection structures are discussed in Section 5.8.

2.5 Data Freeze

All data were frozen prior to analysis. SHA-256 fingerprints were computed for each file and stored in a manifest. All reported analyses use this frozen dataset.

The frozen block table contained 11,764 original rows. We first removed 100 blocks with missing required paired outcome or session metadata. The fewer-than-25-retained-block session rule then removed a further 57 blocks. The final trial dataset therefore comprised 11,607 blocks across 387 retained sessions (Human: 4,737 blocks across 158 sessions; AI: 3,780 across 126 sessions; Baseline: 3,090 across 103 sessions). Of these sessions, 386 contained 30 retained blocks and one Human session contained 27.

Separately, 1,891 audit calls were available across 389 sessions containing audit records. The 389 audit-containing sessions and the 387 retained trial-data

sessions are different counts because the audit inventory was not restricted by the trial-data inclusion funnel.

2.6 Analysis Taxonomy and Stream-Level Metrics

The following metrics were prespecified and computed live for both the Subject and PCS streams during data collection: hit z-score and p-value, cumulative range, the single-scale H_{RS} score, lag-1 autocorrelation, and block-level Shannon entropy (computed over the whole block and separately over halves and thirds). The single-scale H_{RS} score became the main worked example because it could examine ordering within the short streams and produced a pattern that warranted closer examination (Section 1.4). Additional ordering and complexity measures including run count, run-length entropy, higher-order n-gram entropy, and mutual information, were reconstructed from the frozen raw bits and examined as part of the exploratory sweep, testing the analytical range of the architecture and identifying which patterns or limitations warranted attention in Exp 5. The run-count, lag-1 autocorrelation, and run-length entropy measures proved mechanically redundant. They amounted to one measurement under three names, and were treated as non-independent rather than pooled as separate confirmations.

2.6.1 Analysis Taxonomy

As introduced in Sections 1.2–1.4, the Paired-Delta Protocol organizes analysis along two independent dimensions: the analysis family (single-stream, paired

differential, or relational) and the property measured within it (location, dispersion, composition, temporal structure, trajectory, or joint-distribution shape). Table 1 lays out this two-dimensional taxonomy. A given study may populate multiple cells rather than selecting a single one. Its cells represent questions the architecture can support rather than a battery that should be applied in full.

Table 1. Two-Dimensional Analysis Taxonomy for the Paired-Delta Protocol

	Single-stream analysis	Paired differential analysis	Relational analysis
What the analysis family adds	Conventional stream-level measurement	Within-call internal reference and conditional common-mode rejection	Coupling, alignment, stability, lag structure, and joint shape
Location	Mean level in Subject or PCS	Mean Subject-minus-PCS difference	Average association or coupling
Dispersion	Variability within one stream	Variability of the paired difference	Variability of contemporaneous or lagged coupling
Composition	Zero/one balance in one stream	Difference in bit balance between Subject and PCS	Joint bit combinations or same-position co-occurrence
Sequential Structure	Persistence, runs, autocorrelation, n-gram patterns, ordering complexity	Difference in the same ordering measure	Local, lagged, or directional cross-stream dependence

Spectral structure	Periodicity, oscillations, frequency power, spectral entropy	Between-stream difference in spectral features	Cross-spectrum, coherence, phase, shared periodicity
Trajectory	Change across blocks or sessions	Change in the paired difference	Change in coupling over time
Joint-distribution Shape	Not applicable	Not applicable	Nonlinear dependence, tails, clusters, copula shape

Table 1. The columns distinguish the relationship being analyzed, whereas the rows distinguish the property measured. The entries are examples of analytical questions supported by the architecture, not a required battery, and not every listed analysis was implemented in the pilot. Some analyses span more than one property. For instance, the windowed relational-correlation-variance analysis (Section 3.7.2) is classified under Dispersion, but because it is computed across temporal windows it also touches Trajectory.

The taxonomy can be applied either prospectively or exploratorily, as described in Section 1.4.

2.6.2 Hit rate

The proportion of Subject bits matching the intended direction (hits / 150). Within each QRNG call, the paired-delta hit rate was computed as:

$$\Delta_{rate} = rate_{Subject} - rate_{PCS}$$

As with other paired-delta comparisons (Section 1.2), this subtraction cancels contributions expressed equally in both streams' hit rates while retaining contributions that affect them unequally. A surviving difference indicates asymmetry between the labeled streams rather than deviation from an assumed ideal baseline of 0.5.

2.6.3 Single-Scale H_{RS} Ordering Score

The single-scale H_{RS} ordering score was implemented in the experiment's JavaScript source as `hurstApprox` before the pilot data were analyzed. It was used as an exploratory, ordering-sensitive screening measure.

Rescaled-range analysis is classically used to estimate the Hurst exponent from how R/S changes across multiple sequence lengths. Because the implemented function evaluates only the complete 150-bit block, it does not provide a multi-scale estimate of the Hurst exponent. Instead, it produces a single-scale score, denoted H_{RS} , which was used here as an exploratory measure of within-block ordering.

Raw bits were first mapped to -1 and +1. For each 150-bit stream, each value's deviation from the stream mean was accumulated in sequence. R was the difference between the highest and lowest points reached by this cumulative-deviation trajectory, and S was the standard deviation of the 150 mapped values, calculated using n in the denominator. The score, denoted H_{RS} , was calculated as

$$H_{RS} = \frac{\log(R/S)}{\log(n)}, n = 150$$

Values were constrained to $[0, 1]$, consistent with the limits of the Hurst exponent that the function was originally intended to approximate.

R is the range of the cumulative-deviation trajectory, not the range of the raw bit values. Because this trajectory depends on the order in which the bits occur, rearranging the same 150 values generally changes R and therefore H_{RS} . By contrast, S and hit rate remain unchanged when the same bits are rearranged. The score can therefore detect aspects of ordering that are not captured by hit rate alone, although changing the overall bit composition may also affect the score.

Higher H_{RS} values indicate more extended cumulative excursions, while lower values indicate more constrained excursions. Different sequences can produce the same value, so the score does not uniquely identify the ordering pattern responsible for a result. Because it is calculated at only one sequence length, it is not interpreted as a Hurst exponent or as evidence of long-range dependence. The shuffle analysis in Section 3.6.3 and the simulations in Section 5.10 characterize its response under the conditions examined.

2.7 Statistical Framework

Hit rates and single-scale H_{RS} scores have different distributional properties and were analyzed accordingly.

2.7.1 Hit rate analysis

The Bayesian analysis modeled each block's paired target-aligned hit-rate difference, calculated as Subject minus PCS, as a continuous outcome. Fixed effects were included for Human 5+ and AI relative to Baseline and for centered within-session block position. Random intercepts for database identifier and session represented the grouping structure available in the stored data. The 20 database identifiers for the Human 5+ subgroup included Human contributors and accounts used for the automated AI and Baseline conditions; they do not represent 20 independent contributors or fully encode the automated collection-burst structure.

Weakly informative priors were used for all model parameters. The model was implemented in PyMC and sampled using NUTS with four chains, 2,000 tuning draws, and 2,000 retained draws per chain. Results are reported as posterior means and 95% highest-density intervals in Section 3.5.

Hit rate was included primarily to give participants a concrete task during each session and to confirm that block generation and scoring were functioning correctly. The pilot was not large enough to reliably detect the very small hit-rate effects reported in this literature. The weighted mean deviation from chance for the original Radin and Nelson (1989) meta-analysis is 0.00018, as reported in Bösch, Steinkamp, and Boller (2006, p. 501). A post hoc sensitivity calculation indicated that detecting an effect of this size would require approximately 51,320 participants for 80% power or 68,702 participants for 90% power. This calculation was used

only to place the precision of the hit-rate results in context and was not part of the Bayesian model.

2.7.2 Paired Single-Scale H_{RS} score difference (ΔH_{RS})

The single-scale H_{RS} score has substantial finite-sample variability at $n = 150$. The primary uncertainty analyses preserved the observed hierarchy by resampling Human contributors and then whole sessions within contributors, while resampling Baseline sessions as whole units, conditional on the two machines and collection periods represented in the observed Baseline data. Sensitivity analyses compared block-weighted and equal-contributor/equal-session estimates. Equal-session weighting was also calculated but was identical to block weighting because every retained session contained 30 blocks. Whole-session recurrence and leave-one-whole-session-out analyses assessed whether the Human 5+ pattern was recurrent or concentrated in particular sessions. A composition-conditional order-randomization analysis calibrated selection of the largest positive contrast across the documented 2+, 3+, and 5+ thresholds while preserving each stream's observed bit composition. Block-level KS and t-tests are retained only as legacy descriptive calculations and are not used for participant- or population-level inference.

Paired comparisons were computed as:

$$\Delta H_{RS} = H_{RS, Subject} - H_{RS, PCS}$$

The analysis did not apply an Anis–Lloyd correction to the individual scores. Because this correction is a fixed function of sequence length, it would add the same value to both 150-bit scores and therefore cancel exactly in their paired difference. This algebraic cancellation applies only to that equal additive correction; it does not establish that the estimator behaves identically when Subject and PCS have different generating processes or ordering structures. The simulations reported in Section 5.10 examine this possibility. The primary experimental contrasts therefore used paired score differences unless otherwise noted.

2.7.3 Relational analysis framework

Relational analyses examined the association between the Subject and PCS streams.

The pooled relational battery included Pearson correlation between Subject and PCS stream-level scores and same-position mutual information between the two 150-bit sequences. The mutual-information analysis used a circular-shift comparison. The PCS sequence was rotated through its nonzero circular offsets before mutual information was recalculated. This preserved each stream's complete sequence while breaking the recorded same-position alignment between the streams.

Additional analyses examined whether the relationship between the streams varied across local windows, sessions, conditions, or specified temporal offsets. These

analyses asked whether the relationship was stable, rather than only whether the average relationship differed.

The grouping structure used for relational analyses followed the same logic as the other analyses in this paper. Human relational statistics were resampled by contributor because contributors could participate repeatedly. Baseline and AI relational statistics were resampled by session because neither condition contained a comparable repeated-contributor structure. A sensitivity check that grouped Baseline and AI sessions by continuous collection bursts did not indicate that session-level resampling understated uncertainty.

The windowed-correlation-variance and lagged-coupling analyses in Section 3.7.2 were added after the pooled relational battery was complete. They are therefore reported as exploratory extensions rather than prespecified tests.

2.7.4 Interpretation of p-values

The subgroup and diagnostic analyses were exploratory. Their p-values are reported as measures of how compatible the observed results are with the relevant null model. They are not used alone to classify findings as proven or disproven.

The 2+, 3+, and 5+ groups were nested and therefore did not represent independent comparisons. Their block-level KS results are reported descriptively. A composition-conditional order-randomization analysis evaluated both the fixed 5+ contrast and the largest positive contrast across the three documented thresholds while preserving each stream's observed bit composition. This analysis

addresses selection among those three thresholds only. It does not adjust for the wider historical exploratory search. Confirmatory conclusions require a preregistered replication.

2.8 Unit of Analysis

Block-level analyses describe structural differences within the observed data. QRNG calls were generated independently, and the task contains no adaptive stimulus-response channel. However, blocks contributed by the same participant cannot be treated as statistically independent observations for participant- or population-level inference. This dependence is reflected in the clustered bootstrap reported in Section 3.7.1, which produces wider uncertainty intervals.

The paired Subject-minus-PCS contrast is a partial exception to this concern. Because Subject and PCS are drawn from the same 301-bit call, the paired comparison is computed entirely within a single draw and does not require aggregation across participants, sessions, or blocks to be defined. This within-call construction is not subject to between-unit clustering, unlike the cross-group comparisons this paper also reports (Human versus Baseline, condition-versus-condition contrasts, and any analysis that aggregates the paired statistic across repeated observations from the same contributor). Clustering remains a concern for every cross-group and repeated-measures use of the paired statistic; it applies specifically to the single within-call comparison Subject and PCS support, not to the architecture's other analyses.

For the Human 5+ analysis, the estimand is the difference between Human 5+ and Baseline in the mean within-block paired score, $\Delta H_{RS} = H_{RS,Subject} - H_{RS,PCS}$. This contrast was summarized using block-weighted and equal-contributor/equal-session estimates. An equal-session estimate was also calculated, but because every retained session contained 30 blocks, it was identical to the block-weighted estimate and did not constitute a distinct sensitivity result. The equal-contributor/equal-session estimate gave each Human 5+ contributor equal weight and each session within a contributor equal weight.

The Human 5+ subgroup contains 840 blocks organized within 28 sessions from three retrospectively selected contributors. The blocks permit detailed description of the observed sequences, but they are not 840 independent observations for participant- or population-level inference. Hierarchy-preserving bootstrap intervals and session-level influence analyses retain the observed contributor and session structure.

Accordingly, the Human 5+ estimates describe the observed pattern and its sensitivity to weighting within these three contributors. They do not estimate population variability or establish generalizability to new participants.

Whole-session recurrence, leave-one-whole-session-out influence, and a numerical pooled-versus-leave-PI-out comparison are reported in Section 3.7.1.

2.9 Exploratory Human 5+ Session Subgroup

The Human 5+ subgroup comprised the three participants who completed at least five sessions, contributing 28 sessions and approximately 840 blocks. The threshold was examined retrospectively and was initially motivated by the repeated-session recruitment described in Section 2.1. The additional 2+ and 3+ session-count thresholds were examined as follow-up comparisons and are reported in Section 3.6.4.2.

The Human 5+ pattern was summarized using block-weighted and equal-contributor/equal-session estimates. Equal-session weighting was identical to block weighting because every retained session contained 30 blocks. The distinct comparison therefore assessed whether the result changed when contributors, rather than their accumulated blocks or sessions, received equal weight. The subgroup's recruitment and interpretive limitations are addressed in Sections 5.2 and 5.5.

2.10 Computational Environment

Analyses were conducted in Python 3.12.13 using NumPy 2.0.2, pandas 2.2.2, and SciPy 1.16.3. The full analysis pipeline and frozen dataset are available in the public repository (link provided in Data Availability). The analyses extracted hit counts and raw bit arrays from the frozen dataset. Relevant structural measures were then independently recomputed from the raw bits rather than imported from the application's stored summary fields. These included within-block lag-1

autocorrelation and the cumulative-deviation range used to calculate the single-scale H_{RS} score. Within-block lag-1 autocorrelation was analyzed directly, whereas cumulative range was used as a component of the single-scale H_{RS} score rather than as a separate outcome. The original application variables are documented in the repository variable schema.

3. Results

3.1 PCS Calibration Under the Observed Conditions

The Paired Control Stream (PCS) serves as the internal reference for the experimental design. Across all conditions, its hit rate was 0.500253 (session-bootstrap 95% CI [0.499504, 0.501022]; sign-flip $p = 0.5160$).

No condition-related difference was detected in mean PCS hit rate ($p = .7769$) or mean single-scale H_{RS} score ($p = .4696$). Additional hierarchy-aware checks found no clear evidence of across-block lag-1 dependence, unusual run structure, within-session hit-rate drift, or abnormal hit-count dispersion. These analyses calibrate selected PCS properties under the observed conditions with the descriptive distributions and session-level estimates shown in Figure 3.

Figure 3. PCS measures across conditions.

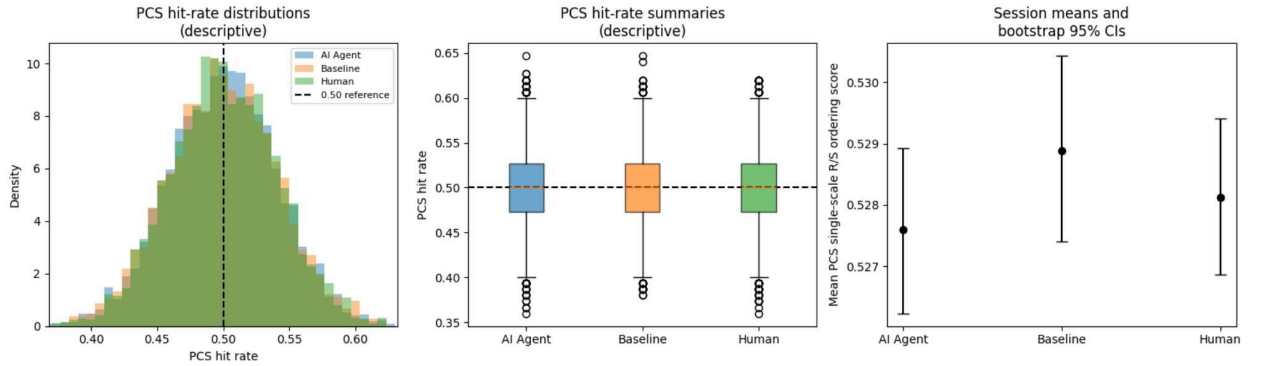


Figure 3. The first two panels show the observed block-level PCS hit-rate distributions and summaries across Human, AI, and Baseline conditions. The third panel shows mean session-level PCS single-scale H_{RS} scores with bootstrap 95% confidence intervals. No condition difference was detected for mean PCS hit rate ($p = .7769$) or mean PCS H_{RS} score ($p = .4696$). These are difference-detection analyses and do not establish exact equivalence or PCS randomness.

These analyses evaluate the PCS alone. We therefore examined the relationship between the two streams directly, using within-call and cross-call comparisons.

3.2 Relational Structure and Paired-Difference Behavior Under Baseline

A potential concern with the paired-delta design is that the two streams derived from a single QRNG call could share common-mode structure. If so, examining the PCS in isolation (Section 3.1) would not characterize the

behavior of the paired difference itself. We therefore examined selected properties of the paired difference using Baseline-condition blocks only ($n = 3,090$), in which no human- or AI-intention task was applied. These analyses describe the joint behavior of the two streams under the Baseline QRNG and processing pipeline. Section 3.2.5 then extends this examination to compare the Subject–PCS relationship across all three conditions.

3.2.1 Common-mode correlation

Shared linear structure in consecutive halves of a single call could produce a positive association between the Subject and PCS scores across Baseline blocks. No such association was detected at the resolution of these analyses. For the single-scale H_{RS} score, Pearson $r = +0.0037$, with a whole-session bootstrap 95% confidence interval of $[-0.0306, +0.0371]$ and a within-session permutation $p = .8357$. The corresponding Spearman correlation was $\rho = +0.0027$ ($p = .8791$). For target-aligned hit rate, Pearson $r = +0.0149$, with a 95% confidence interval of $[-0.0248, +0.0551]$ and a within-session permutation $p = .3966$. These results show that no same-call linear or rank association was detected in the 3,090 Baseline blocks; they do not establish complete independence or exclude weaker, nonlinear, cross-lagged, or time-varying relationships.

3.2.2 Empirical Baseline centering and physical-half symmetry

Across the 3,090 Baseline blocks from 103 sessions, the mean paired single-scale H_{RS} score difference was -0.0017 , with a whole-session bootstrap 95% confidence interval of $[-0.0039, +0.0004]$. The observed distribution had little skew (-0.05) or excess kurtosis (-0.11). The mean paired target-aligned hit-rate difference was $+0.0002$, with a whole-session bootstrap 95% confidence interval of $[-0.0018, +0.0022]$. Thus, neither analysis detected a systematic Baseline difference between the Subject and PCS streams.

Because the implemented protocol divides each call into two sequential 150-bit physical halves before assigning the Subject and PCS labels, the physical positions were also compared directly. The mean physical A-minus-B H_{RS} -score difference was -0.00068 , with a whole-session bootstrap 95% confidence interval of $[-0.00301, +0.00172]$. No systematic physical-half asymmetry was detected at the implemented sequence length. These results do not rule out a boundary or position effect.

3.2.3 Same-call alignment versus within-session cross-call re-pairing

To examine whether the actual within-call alignment contributed detectable linear covariance, each observed Subject score was compared with its PCS score from the same call and with PCS scores reassigned from other blocks within the same session. Reassigning PCS blocks preserved the session membership and the

separate Subject and PCS distributions while breaking their original call-level alignment. The analysis used 10,000 within-session permutations.

The comparison was expressed through the variance of the paired difference. For the single-scale H_{RS} score, the observed paired-delta variance was 0.9963 times the sum of the separate Subject and PCS variances. This ratio fell within the 95% permutation reference interval of [0.9624, 1.0321]. For target-aligned hit rate, the corresponding ratio was 0.9851, within a reference interval of [0.9653, 1.0341]. Thus, preserving the actual within-call alignment did not produce a resolved covariance contribution relative to within-session cross-call re-pairing.

This is a relational diagnostic, but it is not statistically independent of the Pearson analysis in Section 3.2.1. The paired-difference variance is algebraically determined by the component variances and their covariance. Accordingly, this comparison shows the same covariance information through the effect of call-level alignment on paired-delta variance; it does not provide a second confirmation of absent association.

3.2.4 Interpretation

The Baseline analyses detected no same-call Pearson or rank association at the resolution of the available data. Consistently, preserving the observed within-call Subject–PCS alignment did not produce a resolved covariance contribution relative to within-session cross-call re-pairing. These are two representations of the same underlying covariance relationship, subject to the limitations noted above.

These analyses illustrate the relational class of measure introduced in Section 1.5. The correlation and within-call versus cross-call comparisons evaluate Subject and PCS jointly rather than examining either stream alone. Section 3.2.5 extends this approach using additional structure and dependence checks.

These relational analyses do not resolve the separate estimator-bias issue discussed in Section 5.10. In particular, they do not establish that the finite-sample bias of the single-scale H_{RS} score would remain matched if Subject and PCS had different underlying ordering structures. Section 5.10 examines the behavior of the paired score under the specific types of ordering difference generated in that simulation.

3.2.5 Additional structure and dependence checks

Additional analyses examined lag-1 autocorrelation, recurring bit patterns, entropy, and several aspects of the relationship between the Subject and PCS streams. The temporal-structure analyses found no clear condition differences in the features tested.

Mutual information calculated directly from same-position raw bits averaged approximately 0.0047 bits. Same-position mutual information, evaluated against the circular-shift reference described in Section 2.7.3, showed a resolved negative excess for Human at the contributor level ($mean = -0.000262$, 95% $CI [-0.000438, -0.000078]$, $p = 0.0058$, $n = 121$ contributors). Baseline and AI estimates were smaller and individually unresolved: Baseline mean

$= -0.000143$, 95% *CI* $[-0.000365, +0.000083]$, $p = .215$, $n = 103$ sessions; AI mean $= -0.000134$, 95% *CI* $[-0.000330, +0.000066]$, $p = .189$, $n = 126$ sessions. Direct contrasts did not resolve a difference between Human and Baseline ($p = 0.98$) or between AI and Baseline ($p = 0.997$). Because the condition contrasts were unresolved, this pattern does not establish a Human-specific mechanism and is treated as an open, low-priority relational observation.

Using contributor-clustered bootstrap for Human and session-clustered bootstrap for Baseline and AI, the condition-comparison analyses found no resolved differences in Subject–PCS correlation, the variance of the paired relationship, or the joint two-dimensional distribution. Fisher-z correlation contrasts were $+0.0072$, 95% *CI* $[-0.0397, +0.0519]$, for Human minus Baseline; $+0.0061$, 95% *CI* $[-0.0394, +0.0508]$, for AI minus Baseline; and $+0.0011$, 95% *CI* $[-0.0424, +0.0431]$, for Human minus AI. All three pairwise variance ratios were between 0.927 and 1.055, and all energy-distance comparisons had $p \geq 0.19$. An extended battery of lag-1 autocorrelation, n-gram entropy, and entropy-rate slope, computed in direct and relational forms across all three conditions, produced unresolved contrasts in 47 of 48 comparisons. The remaining nominal result sat at the boundary of its interval and was consistent with chance within an unadjusted 48-comparison family. The analyses therefore provide no clear support that the broader Subject–PCS relationship differed between conditions at the resolutions tested.

3.3 Artifact Injection Test

Because the observed Baseline data did not contain a disturbance of known type and magnitude, we evaluated the protocol using implemented artifact-injection controls applied to the raw bits from the 3,090 retained Baseline calls (Section 3.2). Before introducing any perturbation, the raw physical halves were reconstructed and passed through the recorded Subject–PCS assignment and scoring pipeline; all reconstructed scores matched the frozen data.

Two types of synthetic disturbance were then introduced into the physical halves before Subject–PCS assignment: a shared bias in literal-bit composition and increased within-stream persistence. Each disturbance was tested across a range of strengths under two implementations. In the synchronized version, the same positions were affected in both physical halves. In the independent-position version, the two halves received the same type and strength of disturbance but at independently selected positions. Every nonzero level was repeated across 50 independently generated artifact realizations.

After each injection, the complete assignment and scoring pipeline was rerun. Changes in the Subject stream, PCS stream, and their paired difference were measured relative to their original values. Literal-bit results were also examined separately for target-0 and target-1 blocks because a shift toward literal zero or one can have opposite consequences depending on the assigned target.

Uncertainty arising from the simulated artifact realizations was reported separately from uncertainty associated with the sampled Baseline sessions.

Two additional controls tested the boundaries of cancellation. A target-relative intervention applied only to the labeled Subject stream tested whether the paired difference retained a stream-specific change. An unequal-half persistence condition tested whether the paired difference retained a disturbance that affected the two physical halves at different strengths. Together, these simulations tested both when the paired subtraction removes an average shared disturbance and when it appropriately does not.

3.3.1 Shared literal-bit composition bias

For each Baseline call, a randomly selected direction toward literal 0 or literal 1 was applied to both physical halves before Subject–PCS assignment. At each dose (0.02, 0.05, 0.10, and 0.20), the dose represented the probability that a bit position would be replaced by the selected literal value. Each nonzero dose was evaluated across 50 independently generated artifact realizations.

Two forms of the disturbance were tested. In the synchronized-mask condition, the same positions were selected in both halves. In the independent-position condition, each half received the same direction and dose, but different positions were selected. Because literal-bit shifts have opposite target-aligned consequences for target-0 and target-1 blocks, results were also examined separately by target.

Across both implementations, the mean change in the paired hit-rate difference remained close to zero at every dose. At the largest dose of 0.20, the mean recovery error was (-0.000007) for synchronized masks and (-0.000021) for

independent-position masks; both whole-session bootstrap intervals included zero. The block-level recovery-error RMS nevertheless increased with dose, reaching 0.0257 and 0.0347, respectively. Thus, the shared disturbance produced little systematic shift in the paired difference when averaged across calls, while still adding block-level variability. Independent affected positions produced more residual variability than synchronized positions because equal artifact strength does not guarantee identical changes in the two streams.

3.3.2 Shared persistence disturbance

Increased within-stream persistence was introduced at the raw-bit level by giving selected positions a specified probability of repeating the preceding bit. The disturbance was applied to both physical halves before Subject–PCS assignment, after which the single-scale H_{RS} scores and paired differences were recomputed. Doses of 0.05, 0.10, 0.20, and 0.35 were each evaluated across 50 independently generated artifact realizations.

The same two mask conditions described in Section 3.3.1 were used here. The average change in the paired H_{RS} score difference remained close to zero under both implementations. At the largest dose of 0.35, the mean recovery error was +0.00109 for synchronized masks and +0.00111 for independent-position masks; both whole-session bootstrap intervals included zero. However, block-level recovery-error RMS increased with dose, reaching 0.0687 and 0.0697, respectively. Thus, equal persistence strength produced little systematic shift in the

average paired difference, while still adding substantial block-level variability.

These results apply only to the specific persistence mechanisms tested.

3.3.3 Stream-specific and unequal-half controls

A labeled Subject-only intervention served as an implementation check for the assignment and scoring pipeline. Selected Subject-stream bits were replaced with the block's actual target value, while the PCS stream was left unchanged. Because approximately half of the selected bits were expected already to match the target, the analytically expected increase in Subject hit rate was approximately one-half of the replacement probability. Replacement probabilities of 0.02, 0.05, 0.10, and 0.20 produced mean increases in the paired difference of 0.0100, 0.0249, 0.0500, and 0.1000, respectively, closely matching that expectation. Because PCS was unchanged, its paired-difference contribution was identically zero. This control verifies that a labeled Subject-only change passes through the implemented assignment, target-scoring, and subtraction pipeline as specified.

A separate unequal-half control applied persistence to physical half A at twice the dose applied to physical half B. Unlike the equal-strength shared-persistence simulations, this disturbance produced a small systematic residual in the paired single-scale H_{RS} score difference. At doses of 0.10, 0.20, and 0.35 for half A, the mean recovery errors were +0.00069, +0.00106, and +0.00181, respectively, with whole-session bootstrap intervals excluding zero. The Subject-only control verified the implemented processing path, while the unequal-half persistence simulation

demonstrated a boundary of common-mode rejection: paired subtraction need not cancel disturbances that affect the two physical halves unequally.

3.3.4 Interpretation

Under the simulated shared disturbances, the architecture reduced systematic displacement of the paired difference. Disturbances applied to both physical halves with the same type and average strength produced little systematic displacement of the paired difference, although block-level recovery error increased with disturbance strength. The Subject-only intervention separately verified that the implemented assignment, target-scoring, and subtraction pipeline transmitted a labeled-stream change as analytically expected. The unequal-half persistence simulation produced a residual when the two physical halves were disturbed at different strengths. These results illustrate the intended behavior for the mechanisms, doses, and retained Baseline calls tested here. They demonstrate the protocol's operating behavior under the controlled disturbances examined rather than cancellation of an independently characterized, naturally occurring artifact.

Pairing carries a known variance cost: when the two streams have similar variances and are approximately uncorrelated, the paired difference has about $(\sqrt{2})$ times the standard deviation of either stream alone. This loss of precision can be reduced by collecting additional blocks. A systematic common-mode bias in a single-stream analysis, however, would not disappear with additional data. For disturbances that affect the selected stream statistics equally, the paired

architecture trades some statistical precision for reduced systematic displacement. The present tests do not establish this behavior for uncharacterized, asymmetric, temporal, nonlinear, or provider-dependent disturbances.

The two injection mechanisms differed in how precisely they could rule out a residual near the scale of the Human 5+ contrast. For literal-bit composition bias, the largest tested dose (0.20) produced a 95% confidence interval for the mean recovery error of $[-0.00044, +0.00044]$, comfortably smaller than the observed Human 5+ contrast (+0.00435). For the persistence mechanism, the largest tested dose (0.35) produced a wider interval, $[-0.00020, +0.00238]$, whose upper bound reaches roughly half the observed contrast. This reflects the statistical precision achievable with 50 injected realizations per dose rather than evidence of an actual residual bias specific to persistence. A prospectively powered replication with more injected realizations is needed to narrow this interval enough to more confidently rule out an undetected common-mode residual of this scale.

The same-call PCS also made it possible to examine the relationship between the two streams under the observed Baseline operating conditions. The tested within-call correlation, re-pairing, and dependence diagnostics found no clear same-call relationship at their available sensitivity. These results do not prove that the hardware was artifact-free or that the streams were completely independent; they apply only to the particular forms of association tested under the sampled Baseline conditions.

3.4 QRNG Audit-Sequence Checks

Separate 1,000-bit QRNG audit sequences were scheduled after blocks 5, 10, 15, 20, and 25. Based on completed block records, 1,945 audit opportunities occurred across 390 sessions that reached at least one scheduled audit point. The frozen audit table contains 1,891 audits across 389 sessions; 54 scheduled audits were not recorded. Missing audits were concentrated in the AI condition (47 of 636 opportunities), compared with Human (6 of 794) and Baseline (1 of 515), and all occurred during sessions whose recorded block-level provider was `qrng-race`. Because condition, provider, collection date, and execution regime were partly confounded, the source of this missingness cannot be isolated. Audit availability was therefore not treated as evidence that monitoring coverage was equivalent across conditions.

Of the 1,891 recorded audits, 120 lacked complete stored results for the three-test battery. The Frequency (Monobit), Runs, and Longest Run of Ones subtests were reconstructed from the underlying 1,000-bit audit sequences and validated against the 1,771 audits with complete stored results, yielding zero pass/fail disagreements across all three subtests. The Runs subtest reconstruction uses a denominator that differs from the standard NIST SP 800-22 formula; an initial attempt using the standard NIST formula did not reproduce the stored results, but the implemented (non-standard) formula reproduced all 1,771 stored pass/fail classifications exactly, indicating it matches the formula used at data-collection time. On

this basis the 120 missing results were reconstructed and all 1,891 audits were included in the analyses below. Because the stored Runs calculation differs from the standard NIST formula, these results describe the implemented three-subtest battery rather than validation against the complete NIST SP 800-22 suite.

Among the 1,891 audits, 1,845 passed the complete battery and 46 failed at least one subtest, giving an all-three pass rate of 97.57% (session-cluster bootstrap 95% CI: 96.89%–98.20%). The 46 failures occurred in 46 different sessions.

Table 2. Three-Subtest Audit Results

Measure	Human	AI Agent	Baseline	Overall
Audits	788	589	514	1,891
All-three pass rate	97.84%	97.62%	97.08%	97.57%

Frequency pass rate	98.22%	98.81%	97.86%	98.31%
Runs pass rate	100.00%	100.00%	100.00%	100.00%
Longest Run pass rate	99.11%	98.30%	98.83%	98.78%

Table 2 Whole-session permutation comparisons detected no condition difference in complete audit-battery failure rates after Holm adjustment (all adjusted $p=1.00$). Thus, among the available complete batteries, no condition difference in all-three failure rates was detected at the sensitivity of these three subtests. This comparison does not correct for the condition-associated difference in audit availability.

Failure rates were also summarized descriptively by QRNG provider and collection month. No inferential conclusions were drawn from these summaries because provider, date, and collection regime were partly confounded, and some provider samples were very small.

3.5 Mean-Level Hit-Rate Results

The primary frequentist hit-rate analyses examined the all-Human target-aligned Subject-minus-PCS difference and its contrast with Baseline. AI Agent provided a secondary comparison. Human 5+ contained only three participants and was defined and emphasized through retrospective subgroup analysis, so this comparison is descriptive and exploratory rather than a population-level result. Table 3 also reports the Subject and PCS component means. Point estimates were calculated across the observed blocks, while uncertainty was estimated by resampling complete sessions and, for Human analyses, participants. This preserved the within-call pairing and the principal levels of dependence in the data.

Table 3. Target-aligned hit-rate estimates by condition

Condition	Subject mean, % [95% interval]	PCS mean, % [95% interval]	Subject-PCS , percentage points [95% interval]	Difference from Baseline, percentage points [95% interval]
Baseline	50.0179 [49.8831, 50.1575]	49.9978 [49.8434, 50.1443]	+0.0201 [-0.1743, +0.2265]	Reference
AI Agent	50.0284 [49.9049, 50.1506]	50.0060 [49.8813, 50.1365]	+0.0224 [-0.1328, +0.1758]	+0.0023 [-0.2532, +0.2524]
Human-all participants	50.0386 [49.9096, 50.1532]	50.0585 [49.9183, 50.1898]	-0.0200 [-0.2034, +0.1621]	-0.0400 [-0.3190, +0.2246]
Human 5+ exploratory	50.2238 [49.9841,	50.1365 [49.9035,	+0.0873 [-0.3674,	+0.0672 [-0.4159,

subgroup	50.5911]	50.5141]	+0.5875]	+0.6052]
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Table 3. Subject and PCS means are reported separately, followed by the within-call paired difference (Subject minus PCS) and its contrast with Baseline. Positive paired values indicate a higher target-aligned hit rate in the Subject stream than in the same-call PCS stream. All intervals for the paired differences and their contrasts with Baseline included zero. AI Agent and Human-all showed no resolved paired or Baseline-contrast difference. The Human 5+ row represents three retrospectively selected participants, is descriptive rather than a population-level result, and is discussed further in Section 3.6.

A complementary Bayesian model specifically compared Human 5+, AI, and Baseline. Its estimated paired differences were similarly close to zero: +0.018 percentage points for Baseline (95% HDI: -0.275 to +0.283), +0.019 for AI (-0.196 to +0.221), and +0.060 for Human 5+ (-0.372 to +0.503). The estimated Human 5+ versus AI contrast was +0.041 percentage points (-0.451 to +0.510).

The pilot did not show a clear paired mean hit-rate difference. This does not establish equivalence or exclude smaller effects (see Section 2.7.1 for the corresponding participants-needed sensitivity calculation). A session-clustered sensitivity analysis estimated that the Human-all versus Baseline contrast had approximately 80% power to detect a difference of 0.372 percentage points under the specified location-shift model.

3.6 Exploratory Ordering Analysis: Single-Scale H_{RS} Score

The single-scale H_{RS} score was used as a worked example of how the Paired-Delta architecture can examine ordering within paired streams. At the implemented 150-bit length, the score is sensitive to ordering but is not a validated estimate of a Hurst exponent or long-range dependence. The pilot first examined the full Human cohort and then explored session-count subgroups, including the retrospectively emphasized Human 5+ subgroup. These analyses were exploratory and were used to identify patterns and limitations for prospective testing.

3.6.1 Full Human cohort

The mean paired H_{RS} score difference was +0.00113 in the full Human cohort and -0.00174 in Baseline. The resulting Human-all minus Baseline contrast was +0.00287, with a hierarchy-preserving 95% interval of approximately -0.00001 to +0.00575. Thus, the interval narrowly included zero.

For comparison, an unclustered block-level mean test produced $t = 1.94$, $p = .0521$, while an unclustered KS comparison produced $D = 0.0226$, $p = .2884$. Because these block-level tests treat observations from the same sessions and participants as independent, they are reported only as descriptive sensitivity analyses.

Overall, the full-cohort results were borderline and depended somewhat on the inferential method. They do not provide robust evidence of a population-level

ordering difference, nor do they establish that the Human and Baseline conditions were equivalent.

3.6.2 Session-Count Pattern (Exploratory)

The paired single-scale H_{RS} score varied descriptively across the non-overlapping Human session-count strata.

Table 4. Mean paired ΔH_{RS} relative to the paired-null reference and empirical Baseline

Condition or Group	Blocks	From paired-null reference	From empirical Baseline
Baseline	3,090	-0.00174	Reference
AI Agent	3,780	+0.00094	+0.00268
Human-all	4,737	+0.00113	+0.00287
Human, 1 session	3,327	-0.00004	+0.00170
Human, 2 to 4 sessions	570	+0.00322	+0.00496
Human, 5 + sessions	840	+0.00435	+0.00609

Table 4. Paired ΔH_{RS} is the single-scale H_{RS} score for the Subject stream minus the score for the same-call PCS stream; its paired-null reference is zero. The Human session-count strata are mutually exclusive, and Human-all combines the three strata. The final column subtracts the observed Baseline paired mean of -0.00174 and is descriptive rather than a separate inferential test.

The observed paired means increased across the mutually exclusive Human session-count strata. This progression does not establish an effect of repeated participation because session count was confounded with recruitment source and participant characteristics. The negative realized Baseline mean increased every positive empirical-Baseline contrast by the same absolute amount, but it cannot explain the differences among the Human strata because each was compared with the same Baseline value. For Human 5+, the Baseline value accounted arithmetically for approximately 29% of the observed +0.00609 contrast. The Baseline whole-session bootstrap 95% interval, $[-0.003823, +0.000301]$, included zero, and a separate synthetic-data check did not reproduce a consistent paired-difference offset from the processing pipeline. This check addressed a possible artifact of the processing steps, not the natural variation of the Subject and Baseline samples. That variation is discussed further in Section 3.7.1. Both streams can vary across samples, and the Human 5+ contrast was sensitive to the particular Subject and Baseline data observed in this pilot. It should therefore not be interpreted simply as a stable elevation within Human 5+.

The 5+ threshold was initially of interest because Reddit recruits had been encouraged to complete five to ten sessions. It was one of several retrospectively examined thresholds, with the 2+, 3+, and 5+ analyses reported together in Section 3.6.4.2. The 5+ comparison was the strongest of these exploratory thresholds. A composition-conditional random-order calibration was therefore used to evaluate both the fixed 5+ result and selection of the largest positive contrast

across the documented 2+, 3+, and 5+ thresholds. The result is therefore treated as a screening signal and worked example for the paired analysis rather than as evidence of a generalizable subgroup difference.

The AI Agent paired mean was +0.00094. Its unclustered comparison with Baseline produced $D = 0.0227$, $p = 0.3359$. No AI Agent versus Baseline distributional difference was detected at this resolution.

The descriptive variation in mean paired single-scale H_{RS} score across the Human session-count strata is illustrated in Figure 4.

Figure 4. Mean paired single-scale H_{RS} by condition and Human session-count group

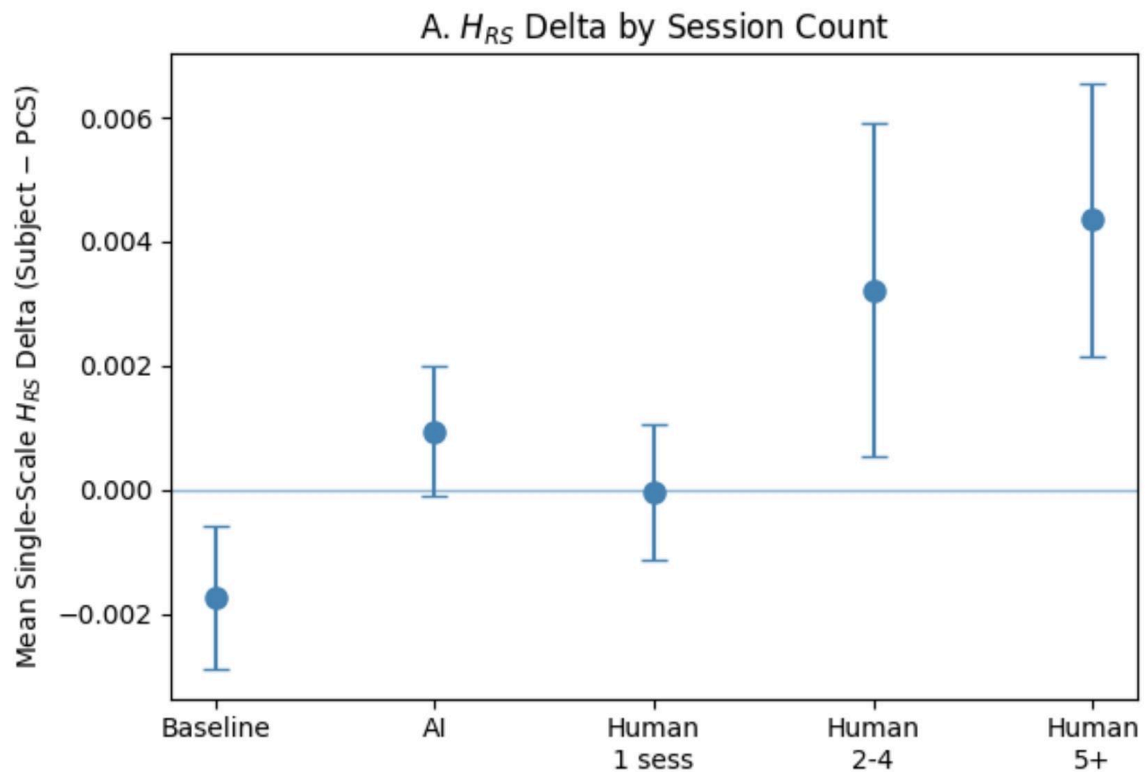


Figure 4. The Human 5+ subgroup had the largest observed mean among the Human strata but contained only three participants and was analyzed retrospectively. Error bars show block-level ± 1 SE and do not represent participant-level uncertainty. The figure is descriptive and does not establish a session-count effect. See Section 3.6.4.2 for the hierarchy-preserving estimate used for inference.

3.6.3 Shuffle Sensitivity to Within-Block Ordering

The H_{RS} score is calculated from the cumulative-deviation path of a sequence, so changing bit order can change the score even when the number of zeros and ones remains fixed. To demonstrate this property, the same positional permutation was applied to both streams within each paired block. This preserved bit composition, pairing, and the shared transformation while changing within-stream ordering.

The analysis used window sizes of 1, 3, 5, 10, 25, 50, and 150 bits. At $W = 1$, the ordering was unchanged. At $W = 150$, each complete 150-bit stream was randomly reordered. The Human 5+ versus Baseline distributional separation changed across the shuffle conditions, with the smallest mean KS distance after full-block shuffling.

This analysis shows that the measured distributional separation changed when within-block ordering was altered while bit composition was preserved, demonstrating that the H_{RS} score responds to ordering at the implemented 150-bit length. The shuffle test characterizes the estimator's sensitivity to ordering, and the Human 5+ comparison itself is evaluated in Section 3.6.4. The full window-specific values and figure are reported in Supplementary Section S3.

3.6.4 Joint Interpretation Framework and Its Application in the pilot

The Paired-Delta Protocol supports two complementary comparisons using the same within-call data (relational analysis, the third analysis family, is addressed

separately in Section 2.7.3). The direct-stream analysis compares the mean Subject-stream M score between Human and Baseline:

$$\bar{M}_{Subject, Human} - \bar{M}_{Subject, Baseline}.$$

The paired-delta analysis first calculates the within-block Subject-minus-PCS difference:

$$\Delta M = M_{Subject} - M_{PCS}$$

and then compares the mean paired difference between Human and Baseline:

$$\Delta \bar{M}_{Human} - \Delta \bar{M}_{Baseline}$$

These analyses address different questions. The direct comparison asks whether the Subject stream differs between conditions. The paired-delta comparison asks whether the relationship between the Subject and PCS streams differs between conditions after removing variation shared within a call. When within-call shared variation is negligible, the direct comparison may be more precise because subtraction introduces variability from both streams. When shared call-level variation is present, subtraction may reduce that variation and provide a more specific test of differential stream behavior.

Because statistical significance depends on both the estimated differences and their uncertainty, the four combinations below do not uniquely determine the positions of the Subject and PCS distributions. They instead provide an interpretive

framework that must be considered together with effect directions, magnitudes, uncertainty intervals, and the relational analyses.

In the pilot, the worked example used the single-scale H_{RS} score. The direct comparison was the difference in Subject-stream H_{RS} between Human and Baseline. The paired comparison was the difference in ΔH_{RS} , where

$\Delta H_{RS} = H_{RS,Subject} - H_{RS,PCS}$. The same framework can be applied to any selected stream-level measure.

Table 5. Interpretive framework for direct-stream and paired-difference results

Direct-stream result	Paired-delta result	Reading
Difference detected	Difference detected	The Subject stream differs between conditions, and the within-call Subject minus PCS relationship also differs. Interpretation depends on the direction and size of both results.
Difference detected	No difference detected	The Subject stream differs between conditions, but no difference in the within-call relationship was detected. The effect may be expressed similarly in both streams, or the paired analysis may have lower sensitivity.
No difference detected	Difference detected	The within-call Subject minus PCS relationship differs between conditions even though no direct Subject-stream difference was detected. This can occur if PCS differs between conditions or if

		pairing removes shared variation.
No difference detected	No difference detected	Neither analysis detected a difference at the available resolution.

Relational statistics are not included in this table because they measure the relationship between the two streams rather than the level of either stream or their within-call difference. They are reported separately in Section 3.2.

The following subsections apply this general framework to the exploratory Human 5+ result from the pilot.

3.6.4.1 Application to the Human 5+ Comparison

For the 5+ session subgroup, the unclustered direct comparison of Subject-stream H_{RS} scores between Human 5+ and Baseline ($H_{RS,Subject,Human}$ vs $H_{RS,Subject,Baseline}$) produced KS $D = 0.0575$, $p = 0.0243$. The corresponding comparison of the paired Subject-minus-PCS scores produced KS $D = 0.0541$, $p = 0.0398$. Both comparisons therefore showed nominal distributional differences in the same interpretive direction, with similar KS distances.

The observed separation remained in the paired comparison rather than appearing only in the direct Subject-stream analysis. Its interpretation requires examination of the separate Subject and PCS contributions in Section 3.6.4.2.

A within-block label-swap analysis produced a similar block-level result, but it evaluates the same paired contrast and is not independent confirmation. The complete label-swap results are reported in Supplementary Section S4.

The observed combination corresponds to the first row of Table 5 and is examined through the stream-level decomposition in Section 3.6.4.2 and the assignment-timing assumptions in Section 5.9.

3.6.4.2 Bidirectional Stream Pattern and Subgroup Selection

The hierarchy-preserving Human 5+–Baseline contrast in the mean paired H_{RS} was +0.006090, with a descriptive 95% bootstrap interval of [+ 0.000776, + 0.016003]. This is the main subgroup estimate because its bootstrap interval accounts for blocks being grouped within sessions and contributors. The separate stream comparisons below are secondary descriptions of how the paired contrast arose.

To understand why the delta was large in this subgroup, the individual stream distributions were examined separately. For the 5+ subgroup, the Subject stream (Human 5+) trends above baseline ($t = + 1.770$, $p = 0.077$) while the PCS (Human 5+) trends below baseline ($t = - 1.668$, $p = 0.096$). The Subject and PCS estimates differed from Baseline by comparable magnitudes but in opposite directions: +0.00315 for Subject and -0.00294 for PCS. These stream-level comparisons and the label-swap sensitivity analysis are unadjusted exploratory results.

The 5+ threshold was examined retrospectively, followed by the additional 2+ and 3+ thresholds. These groups were nested because the same three 5+ participants also contributed to the 2+ and 3+ groups. Complete stream-level and threshold results are reported in Supplementary Section S4.

To account for selecting among the three documented thresholds, an order-randomization analysis was performed while preserving the observed number of zeros and ones in each stream. Within each of 20,000 repetitions, bit positions were randomized, paired H_{RS} scores were recalculated, and Human-versus-Baseline contrasts were obtained. The one-sided tail area was .00695 for the fixed 5+ comparison and .00800 when the largest positive contrast across the 2+, 3+, and 5+ thresholds was evaluated. The latter analysis accounts for selecting the strongest result among these three thresholds. It does not account for the wider exploratory search or provide contributor-level population inference.

This analysis differs from the shuffle sensitivity analysis in Section 3.6.3. The shuffle analysis measured how progressively disrupting the observed sequences changed the original separation. The analysis here instead created a random-order reference distribution with the observed bit composition held fixed and evaluated selection among the three documented thresholds.

The opposed stream directions are inconsistent with the simplest common-mode explanation in which both streams shift equally and the effect cancels under subtraction. However, this pattern can also arise when a subgroup is selected for

having a large Subject-minus-PCS difference. With weak correlation and comparable variability between the streams, selecting a large difference tends to place one stream above and the other below the reference level. The observed split of +0.00315 for Subject and -0.00294 for PCS closely resembles this pattern. The result therefore does not distinguish differential stream behavior from the consequences of selecting a large paired difference.

The Human 5+ subgroup contained 840 blocks but only three participants. The large block count does not capture between-participant variability or support inference beyond these contributors. Section 3.7.1 reports the influence of the principal investigator's contribution on the subgroup estimate, and Section 5.5 discusses experimenter participation. The subgroup result remains exploratory and requires preregistered replication.

3.6.5 Run-structure Check

Run structure was examined to characterize what the single-scale H_{RS} score measures in the observed 150-bit sequences. Using the 1st and 99th percentiles of the Baseline Subject-stream H_{RS} distribution as exploratory in-sample cutoffs, Human blocks were classified into lower-tail, middle, and upper-tail regions.

The two tails showed different run patterns. Lower-tail blocks had more runs and shorter maximum runs than typical blocks, whereas upper-tail blocks had fewer runs and longer maximum runs. Mean run counts were 79.44, 75.49, and 68.51 in the lower-tail, middle, and upper-tail regions, respectively; corresponding mean

maximum runs were 5.67, 7.58, and 9.60 bits. This pattern is expected because longer same-value runs tend to produce larger cumulative excursions, which increase the single-scale H_{RS} score. The relationship is not an algebraic identity, but the measures are structurally related and calculated from the same sequences. The tail comparison therefore illustrates the score's empirical behavior rather than providing a demanding or independent validation.

A separate paired analysis examined whether the Subject-minus-PCS run structure differed between Human 5+ and Baseline while preserving session and participant hierarchy. The Human 5+–Baseline contrasts were –0.058 runs for run count (95% descriptive interval: –0.995 to +0.778), +0.048 bits for maximum run length (–0.145 to +0.279), and +0.00089 for run-length entropy (–0.01379 to +0.01772). All intervals included zero. Thus, the direct run-based metrics did not independently reproduce the Human 5+ paired H_{RS} -score difference at the resolution of this analysis.

Run count was treated as the primary run measure. Maximum run length and run-length entropy were correlated secondary descriptions. The Baseline cutoffs, run metrics, and Human 5+ comparison were exploratory.

3.6.6 Bit Composition

Order-1 Shannon entropy was used as a measure of bit balance. The paired Human 5+ Subject-minus-PCS entropy difference was –0.00036 (95% descriptive interval: –0.00111 to +0.00070). Relative to Baseline, the Human 5+ paired

contrast was -0.00044 (-0.00127 to $+0.00068$). Both intervals included zero, providing no clear evidence that the Human 5+ Subject–PCS H_{RS} -score difference was accompanied by a difference in bit composition. The observed H_{RS} -score pattern was not accompanied by a detected imbalance in the proportion of zeros and ones. This provides no clear support for simple bit-composition imbalance as an explanation, but it does not by itself establish that the pattern was caused by ordering.

The original analysis compared Subject-stream entropy directly across conditions and found no clear condition difference. As an additional relational analysis, entropy was calculated separately for the two streams produced within each call and expressed as Subject minus PCS. Across all Human records, Subject entropy was slightly lower than PCS entropy at n-gram orders 1 through 4, with a larger same-direction difference in Human 5+. AI and Baseline remained near zero and did not show a consistent direction across entropy orders. All hierarchy-preserving intervals included zero. The consistently negative Human 5+ estimates therefore represent a weak same-direction pattern that is compatible with sampling noise and does not establish a difference in entropy or short-pattern structure. The analysis nonetheless illustrates how the paired architecture can examine the relationship between same-call streams even when direct comparisons of the Subject streams do not clearly differ across conditions.

3.6.7 Variance Comparison

The block-level variance of the paired Subject-minus-PCS H_{RS} score was nearly identical in Human 5+ and Baseline. The Human 5+ variance was 0.004091, compared with 0.004116 for Baseline, producing a variance ratio of 0.994 (descriptive 95% interval: 0.865–1.172). No clear increase in paired-score variability accompanied the subgroup mean difference. Finite-sample variability is a general feature of the single-scale H_{RS} score at the implemented 150-bit length, although this particular comparison addresses variability in Human 5+ relative to Baseline rather than estimator bias.

3.6.8 Interpretation

The analyses found no clear support for several simple artifact explanations. Composition differences were not detected in the relevant comparisons (Section 3.6.6) and no increase in paired H_{RS} -score variance was detected (Section 3.6.7). An equal common-mode hardware shift would tend to cancel in the paired difference, while the audit analyses found no clear systematic change in the separate audit calls (Sections 3.2.1 and 3.4). The positive direction also remained under the alternative weighting schemes examined in Section 3.7.1; block and equal-session weighting were identical because all sessions contained the same number of retained blocks.

The Human 5+ result that remains after these checks is a mean shift, reported as the strongest result among several nested session-count thresholds and present in

both the direct and paired analyses. These analyses are computed from the same stream scores and are therefore not independent tests, although they answer different questions (Section 3.6.4). The label-swap permutation is a sensitivity analysis of the same paired contrast rather than independent confirmation, with complete results reported in Supplementary Section S4. The shuffle analysis shows that the measured separation changes when within-block ordering is disrupted, but it does not establish that the original subgroup difference was systematic. Full shuffle results are reported in Supplementary Section S3.

Two qualifications remain particularly relevant. First, the single-scale H_{RS} score has a finite-sample response that depends on the generating process and its ordering parameter. The bounded calibration in Section 5.10 found no resolved directional distortion within the examined AR(1) and fractional-Gaussian-noise models, bounding the estimator's directional asymmetry within those models well below the observed Human 5+ contrast. The bound applies only to the generating models examined. The process underlying the QRNG data remains unknown, and no correction to the observed score is available.

Second, robustness across alternative ordering metrics was limited. The Human 5+ versus Baseline paired contrast was repeated using alternative stream metrics in place of the single-scale H_{RS} Score. Each metric was evaluated both directly on the Subject stream and as a paired Subject-minus-PCS difference using the same contributor-level and session-level clustered bootstrap. Lag-1 autocorrelation showed no resolved Human 5+–Baseline contrast in either form (direct:

-0.001627, 95% CI [- 0.005044, + 0.001501]; relational: +0.000112, 95% CI [- 0.004811, + 0.007279]). N-gram entropy at orders 1 through 4 likewise showed no resolved contrast in either form, and the entropy-rate slope contrasts were unresolved. None of these alternative metrics reproduced the nominal pattern observed with the single-scale H_{RS} score in either the direct or paired analysis. Because the alternative measures capture different aspects of ordering, their results do not disprove the H_{RS} result, but they reinforce its metric-specific and exploratory status.

A cross-check using DFA showed the same directional Human 5+ pattern, but with a smaller standardized difference ($d = 0.065$, compared with $d = 0.095$ for H_{RS}) and a larger unadjusted p-value ($p = .0874$). The DFA result was exploratory and did not independently reproduce the Human 5+ subgroup difference. Its direction suggests that the observed pattern was not confined exclusively to the H_{RS} calculation.

Section 3.6.4 explains why the observed directional configuration does not distinguish a systematic stream-specific difference from retrospective selection, and Section 3.7.1 reports whole-session recurrence and a numerical PI-influence check.

3.7 Additional Robustness Checks

3.7.1 Sensitivity to Position, Weighting, Recurrence, and Session Order

Several additional checks examined whether the Human 5+ pattern depended on within-session position, contributor weighting, individual sessions, or session order, with full results reported in Supplement S1. The prespecified continuous position-slope analyses were unresolved in every condition. The Human 5+–Baseline contrast remained positive under block weighting and equal-contributor/equal-session weighting, and its direction did not depend on the principal investigator or any single session. First-versus-later-session comparisons showed mixed contributor-specific trajectories.

A subsequent post-hoc split-half diagnostic found that the first 15 Baseline blocks had a lower paired-score mean than the final 15 blocks at the unadjusted level. The corresponding early-versus-late contrasts were unresolved in Human and AI, and none of the three condition-specific contrasts remained resolved under the Bonferroni threshold of $0.05/3 = .0167$. The Human 5+–Baseline contrast was $+0.00992$ in the early half and $+0.00226$ in the matched late half, but the direct test of whether the condition contrast changed between halves was unresolved. Comparing all Human 5+ blocks with only the late Baseline blocks also reduced the contrast to $+0.00377$, with a 95% interval of $[-0.00184, +0.01366]$. These diagnostics indicate that the negative early-session Baseline realization contributed to the magnitude of the observed subgroup contrast. However, the unresolved direct comparison between halves provides no clear evidence of a systematic

position effect or pipeline mechanism. All of these analyses remain sensitivity checks conducted on the same three-contributor subgroup rather than independent confirmation.

3.7.2 Relational Capability: Local and Lagged Coupling

Beyond the pooled relational battery in Section 3.2, two further relational analyses illustrate the architecture's capacity to examine how within-call Subject and PCS streams co-vary, rather than testing either stream's level alone. Local Subject–PCS coupling, calculated within nonoverlapping windows of five and ten blocks per session, showed no clear difference in average coupling strength between Human 5+ and Baseline. What differed was the stability of that coupling. Window-level correlations were roughly 17% less variable at $W=5$ and 26% less variable at $W=10$ in Human 5+ than in Baseline. This is a difference in how consistently the two streams tracked each other across a session and it is a question no single-stream analysis can pose. With only three contributors, no resampling method produced reliable confidence intervals for this ratio, so it is reported as a candidate pattern rather than a resolved difference. A qualitatively similar point-estimate pattern appeared in a separately collected dataset with longer sessions, though the two could not be compared at a common window size (Supplement S2.1).

A lagged cross-correlation analysis across lags -3 through $+3$ found one candidate pattern, lower between-session variance in the Subject–PCS correlation at lag $+2$, that remained below the adjusted threshold for the complete 14-comparison lag

family (conservative adjusted $p = 0.01560$) but not the broader family of exploratory analyses conducted across the pilot as a whole.

Both patterns concern the stability of the Subject–PCS association across windows or lags, not its average strength, and neither identifies an underlying mechanism. Both analyses were retrospective and exploratory. They are reported here as illustrations of what relational analysis under this architecture makes askable, not as established condition-specific findings. Full detail, including the sensitivity checks and the mean-comparison results that did not survive correction, is reported in the Supplement Section S2. Confirmatory testing of either pattern is reserved for Exp 5, the preregistered replication (Section 4.4).

4. Discussion

4.1 Architectural Contribution

Section 1.1 identified two distinct problems in this literature, and Section 1.2 defined common-mode rejection as the removal of contributions expressed equally in the two stream statistics. The Paired-Delta Protocol addresses within-study ambiguity in two complementary ways. Paired-differential analysis provides conditional common-mode control and reveals how the two streams contribute to a contrast, while relational analysis makes properties of the pair itself measurable. Conceptually, the paired-control architecture resembles differential measurement strategies used in physical instrumentation, including interferometric designs, in which two channels sharing a common source or measurement environment are

compared so that contributions expressed equally in both are removed while contributions that differ between them are retained. The analogy also extends to relational measurements. Just as interferometric observables can depend on the coherence between two optical paths, the paired-control design can examine not only the difference between Subject and PCS but also how their fluctuations covary locally or across temporal offsets. The analogy is architectural rather than physical. Subject and PCS are partitions of recorded QRNG output and do not constitute interfering quantum pathways. The value of the comparison lies instead in converting an absolute measurement against a theoretical expectation into a within-call differential measurement against a contemporaneous reference. A surviving paired difference therefore establishes an asymmetry between the labeled stream statistics, although it does not by itself identify the cause.

The relational case is worth stating in plain terms, because it is the least familiar of the three analyses and the easiest to conflate with a difference in level. Consider two dancers performing together. Being in step is not a property of either dancer alone, so a second dancer turns the pair itself into something to measure, and that pair-level property can be assessed in many ways. One, for example, may view how in step they are on average, whether that degree of synchrony holds steady or drifts, and whether their most extreme departures from the usual pattern happen together. None of these questions can be asked of a single dancer. The comparison this architecture

supports asks the same thing of each: does this look different for a watched pair than for the same pair rehearsing with no one there.

The architecture does not resolve Problem 1, cross-study credibility. Publication bias, selective reporting, and analytic flexibility require process-level remedies applied before outcomes are known, including preregistration, locked datasets, and transparent reporting. This paper addresses that problem partially through in-code metric prespecification and a cryptographic data freeze; the more complete remedy is Exp 5, the preregistered replication motivated by this pilot (Section 4.4). The architecture strengthens within-study interpretation and replication addresses the credibility and reproducibility of findings across studies.

No common-mode relationship was detected in the Baseline data at the resolutions tested, so cancellation of a naturally occurring disturbance could not be demonstrated directly. Instead, the paired null was evaluated empirically, and injection tests demonstrated the expected behavior, while stream-specific contributions survived with the expected variance cost. The design also enabled the within-call versus cross-call comparison in Section 3.2, which would not be available from a single stream. These results support the architecture under the sampled conditions without establishing that the hardware was universally free of common-mode structure.

The relational contribution does not depend on a common-mode artifact being present. Because each call produces two same-call streams, the architecture

permits analysis of their association, temporal alignment, stability, and joint distribution. These properties have no single-channel equivalent.

A third contribution is metric flexibility. Any appropriately calibrated stream-level statistic computable separately for Subject and PCS can be used in the paired differential comparison. The single-scale H_{RS} score, already implemented in the pilot codebase, served as this study's worked example. A candidate pattern in a small, retrospectively selected subgroup provided a live case study in evaluating such a metric. Further analysis showed that retrospective subgroup selection could produce the observed directional configuration and therefore remained a viable explanation. Separately, the score's finite-sample behavior cannot be generalized beyond the generating models examined. Within the bounded AR(1) and fractional-Gaussian-noise simulations, however, no resolved directional distortion accounted for the full observed contrast. This illustrates the architecture's utility as a testbed for evaluating candidate metrics, diagnosing ambiguous patterns, and defining requirements for preregistered replication.

4.2 What the Pilot Data Suggest

This study was designed primarily to develop and evaluate the Paired-Delta Protocol (PDP) rather than provide a powered confirmatory test of observer-linked deviations from chance. Detection of ordering-sensitive structure was a secondary exploratory objective. The protocol was intentionally brief, with 30 trials per session

and 150-bit blocks, prioritizing accessibility over statistical resolution. No clear paired mean hit-rate difference was detected at the resolution of the pilot.

A candidate difference in the single-scale H_{RS} score emerged in the Human 5+ subgroup and served as a worked example of the protocol's diagnostic value. The subgroup consisted of three contributors and was analyzed retrospectively.

Although the 5+ threshold was initially motivated by repeated-session recruitment, session count was an imperfect proxy for recruitment source, additional thresholds were examined, and the strongest result was emphasized. The observed contrast therefore remained sensitive to subgroup definition and selection.

The paired architecture revealed that the candidate result was more complex than the conventional Subject-stream comparison suggested. The Human 5+ Subject stream was +0.00315 above its Baseline counterpart, while the Human 5+ PCS was -0.00294 below its Baseline counterpart. The larger paired contrast was therefore formed about equally by movement in opposite directions across the two streams rather than by an isolated upward Subject-stream difference. Given the streams' similar variability and weak association, this configuration was compatible with what can arise when selecting a large paired difference. It does not establish that selection caused the result, but it shows why the Subject and PCS directions cannot be treated as independent evidence of differential behavior. This distinction could not have been evaluated from the Subject stream alone.

The paired Baseline provided an additional level of resolution. Its negative Subject-minus-PCS realization contributed approximately 29% of the Human 5+ versus Baseline contrast. The observed difference was therefore enlarged by internal behavior of the particular comparator sample rather than arising solely from the Human Subject stream. The concentration of the negative Baseline difference in the early portion of sessions, together with its attenuation under retrospective interleaving, further showed sensitivity to the contiguous stream partition. This is consistent with a position-related contribution, but it does not establish that the original pattern was artifactual, explain why it appeared specifically in Baseline, or show that interleaving is generally superior.

Metric calibration identified another limitation of interpretation. Within the generating models examined, the single-scale H_{RS} score could make genuine differences in temporal structure appear smaller when calculated from short blocks. The simulations did not show that this behavior could create a paired difference when no underlying difference existed, and no resolved directional distortion accounted for the full observed contrast. These results narrow the demonstrated estimator concern without establishing its behavior under untested ordering structures.

Taken together, the analyses did not determine whether the Human 5+ pattern reflected reproducible differential behavior in this population, retrospective selection, the particular Baseline realization, sampling variation, unmodeled metric behavior, or some combination of these factors. Their methodological value lies in

showing that the direct and paired findings did not provide independent support for a single interpretation and that the candidate contrast had identifiable Subject, PCS, and Baseline components. PDP therefore provided a more complete and appropriately qualified account than the conventional Subject stream alone. This is the diagnostic leverage needed in a field where small, ambiguous deviations can otherwise appear to provide stronger evidence than they do.

4.3 Analytical Value of the Three Analysis Families

The analytical value of PDP lies in three complementary views of the same physical draw. Direct-stream analysis retains the conventional comparison between the Subject stream and an external reference. Paired-differential analysis shows how Subject and the same-call PCS jointly produce a within-call contrast. Relational analysis examines properties of the pair that are not defined for either stream alone. These are complementary analyses rather than independent opportunities to search for significance.

Direct and paired analyses answer different questions. The direct comparison asks whether the Subject stream differs between conditions. It avoids the additional variability introduced by subtraction but lacks a contemporaneous within-call reference. The paired comparison asks whether the Subject-minus-PCS relationship differs between conditions after contributions expressed equally in the two stream statistics have canceled. When shared variation is negligible, the direct analysis may be more precise. When a contribution shifts both statistics equally, subtraction can reduce its systematic influence. The controlled artifact-injection

tests demonstrated this operating behavior for the specified disturbances examined here rather than establishing general protection against artifacts.

Applying both analyses also reveals how an observed contrast is assembled. A paired difference may arise primarily from Subject, primarily from PCS, from unequal movement in the same direction, or from movement in opposite directions. Similar aggregate differences can therefore have different component configurations and remain compatible with different explanations. Interpretation requires consideration of the component directions, magnitudes, uncertainty, sampling hierarchy, and selection process rather than merely whether the paired result reaches a statistical threshold.

A third class of analysis is available and is different in kind from the first two. Hit rate and the single-scale H_{RS} score are properties of a stream. The association between the two streams is a property of the pair. Relational measures are computed on Subject and PCS together. Zero-lag correlation, covariance, and mutual information are symmetric under exchange of the stream labels. Directional measures, including signed lagged cross-correlation, lead–lag relationships, and predictive dependence, retain information about which stream occupies each role. Relational analysis therefore does not always require predicting which stream will differ, but its label dependence is determined by the selected statistic. These measures have no single-channel equivalent because a relationship requires two jointly observed series.

The idea that an observer-related effect might not appear as a stable shift in any single stream, but rather in the relationship between two streams, has a precedent in von Lucadou's Correlation Matrix Method, which considers whether such differences may be distributed across correlation patterns rather than concentrated in one outcome variable. The PDP method differs substantially in structure and scope. The Correlation Matrix Method examines correlations across many psychological and physical variables collected across separate trials, while the relational analysis introduced here compares the windowed correlation and joint distribution between two physical halves of a single QRNG call across conditions, testing whether that within-call relational structure differs between Human and Baseline sessions rather than describing it in isolation. What the two methods share is the premise that a relational or distributed signature may be a more natural place to look than a shift in a single marginal statistic. Von Lucadou further proposed that control data traditionally treated as external to the experiment might instead be part of its operational closure, and thus part of the system as a whole (Grote, 2017). Walach raises a related possibility from his own data, suggesting the effect "may be embedded within the whole experiment and not only within the experimental matrix" (Walach, 2021). If this framing has merit, it points to one structural difference worth noting between conventional designs and PDP. Control data is traditionally generated in a separate batch, at a different moment, with no shared history with the subject data it is meant to contextualize. If organizational closure matters the way von Lucadou and Walach suggest, a relational effect distributed across that closure may be easier to see from inside the system, where

subject and control share a history, than from outside it. The Paired-Delta architecture responds to this structurally with the Subject and PCS streams originating from the same call and sharing the same physical history, rather than being drawn from separate acquisitions after the fact.

This precedent motivates why a relational analysis is worth building. We do not know what it will find. PDP is not the Correlation Matrix Method nor does it test the Model of Pragmatic Information. PDP's examination is much more narrow and only asks whether Human and Baseline conditions differ in the variability or temporal alignment of the relationship between two same-call QRNG streams that share a single call's history.

Under Baseline, no Subject–PCS association was detected at the resolutions tested (Section 3.2), supporting the paired reference under the sampled hardware and processing conditions without establishing complete independence.

Exploratory applications under observer conditions are reported in Section 3.7.2.

These analyses examined whether Subject–PCS association varied across local windows and temporal lags. The relational analyses did not identify clear differences in average Subject–PCS coupling. They instead identified exploratory differences in coupling variability: narrower local-window correlation distributions in the Human 5+ subgroup and lower between-session variability of lag +2 cross-correlation in the full Human condition. Because these analyses were retrospective and the mechanisms underlying the variability differences are

unknown, they are treated as exploratory demonstrations of relational questions enabled by the architecture.

The relational class of analysis extends beyond the static, pooled measures reported in Section 3.2. Because the architecture yields two same-call streams from every draw, relational questions extend beyond the analyses pursued here. The local-window analysis examined whether the strength of association varies within a session rather than only across the session as a whole, while the lagged analysis examined cross-block associations between the streams. Other relational questions remain available. For example, the sign of an association may remain stable or shift over time independently of changes in its strength. Measures sensitive to the shape of the joint distribution could also test whether extreme or otherwise distinctive values in one stream coincide with those in the other. These possibilities illustrate that relational analysis is not a single test but a broader class of questions made possible by two within-call streams.

Taken together, the three analysis families provide two forms of added value. Direct and paired-differential analyses reveal how Subject, PCS, and the empirical Baseline contribute to an observed contrast, providing checks on interpretations drawn from the conventional Subject stream alone. Relational analysis makes otherwise inaccessible properties of the pair measurable, including their association, temporal alignment, stability, and joint distribution. The pilot demonstrates both capabilities without classifying the Human 5+ pattern as either an effect or an artifact. It provides a more informative account of the candidate

result while also illustrating a broader class of questions that a single-channel design cannot pose. The prospectively specified replication described in Section 4.4 will test the candidate patterns identified here under conditions designed to address their retrospective provenance.

4.4 Planned Pre-Registered Replication and Permeability Framework

A follow-up study has been provisionally preregistered and will be finalized before data collection. Its primary confirmatory analyses will test the paired differential and corresponding single-stream findings. Secondary relational analyses will examine the pilot finding of reduced variability in local Subject–PCS correlations and explore cross-correlation variability across a prespecified range of temporal lags. The lagged analysis is motivated in part by the candidate lag +2 result observed in the pilot, but no individual lag will be treated as a confirmatory target. The complete lag family will be reported with a prespecified multiplicity adjustment, and any recurring pattern will be considered a candidate for subsequent independent replication.

Because these relational hypotheses were formulated after examining the pilot results, their evaluation in new data will constitute prospective replication, although they will remain secondary to the primary confirmatory analyses.

The study will also explore individual differences in provisional “permeability,” defined as reduced filtering at the boundary between the person and available information. Candidate predictors include psi belief, self-assessed ability, anomalous experience, meditation practice, the MAIA-2, and the BQ-18, with the

HSP-12 and Revised Transliminality Scale added if permission is obtained. These analyses are exploratory and are intended to inform possible future development of a permeability measure.

Before data collection, the populations, estimands, statistics, directions, sampling hierarchies, resampling procedures, and multiplicity adjustments for all replication tests will be locked. The final estimator, bias-control procedure, power analysis, and sample-size justification remain subject to statistical review. Potential responses to the finite-sample behavior identified in Section 5.10 include calibration, alternative estimators, longer blocks, and simulation-based reference distributions.

A retrospective reconstruction of the Baseline calls also informed the follow-up design. Distributing both streams across the full acquisition interval through locally balanced interleaving substantially attenuated the session-position pattern observed under the original contiguous split. This finding does not establish that the original pattern was artifactual or that interleaving is generally superior, particularly because the alternative partition was evaluated in the same data that revealed the pattern. Interleaving is not part of the current Exp 5 design as it is a replication of the pilot. Whether to adopt it remains an open question for future work.

Longer streams are also a candidate for future work. Increasing stream length could improve the finite-sample behavior of the single-scale H_{RS} score and lower

the detection floor for mutual information, although the magnitude of either improvement has not been established using genuine longer sequences.

Link to Exp 5 Pre-registration:

(https://osf.io/yseqj/overview?view_only=98fef7fec86244749d80e778424a065f)

4.5 The Single-Scale H_{RS} Ordering Score as a Metric

The single-scale H_{RS} score served as an exploratory ordering-sensitive screening statistic (Section 2.6.3). Run-structure analysis helped characterize what the score detects. Blocks with lower H_{RS} values generally contained more runs and shorter maximum runs, whereas blocks with higher values had fewer runs and longer maximum runs. Longer same-value runs tend to produce the larger cumulative excursions measured by the single-scale H_{RS} score, so this association was expected rather than an independent validation of it. Paired run-count, maximum-run-length, and run-length-entropy contrasts did not reproduce the Human 5+–Baseline H_{RS} contrast at the available resolution (Section 3.6.5). H_{RS} therefore appears to summarize aspects of ordering that were not captured fully by any one of these simpler measures.

The single-scale H_{RS} score provides useful sensitivity to small ordering differences that may not be expressed as changes in hit rate. In the candidate-scale injection analysis, the pipeline recovered an imposed persistence difference that produced a paired H_{RS} contrast near the scale observed in the pilot. This demonstrates

sensitivity to the simulated form of ordering structure examined. The score is also versatile within the paired architecture: it can be calculated separately for Subject and PCS, expressed as their paired difference, or used to examine how the two streams' ordering scores vary together. Its sensitivity and interpretation depend on the sequence length and the form of ordering structure being measured, so each application requires appropriate calibration.

Future studies should prospectively specify both a mean-level measure and a block-length-appropriate ordering measure. Hit rate measures aggregate target-aligned bit frequency, whereas H_{RS} serves here as an exploratory flag for within-block ordering. Used together, they distinguish mean-level composition from ordering-sensitive structure, although neither identifies the mechanism responsible for an observed difference.

4.6 Bit Composition, Temporal Ordering, and Relational Structure

The pilot illustrates three distinct properties of short QRNG sequences. Composition analyses found no clear condition differences in bit balance or short-pattern frequencies (Section 3.6.6). The single-scale H_{RS} score instead measures aspects of within-block ordering, while relational measures describe how Subject and PCS vary together. These properties are complementary rather than interchangeable. Stream-level composition and ordering measures can also be expressed as paired Subject-minus-PCS differences, providing conditional rejection of variation expressed equally in both stream statistics.

The exploratory relational analyses identified two different forms of reduced variability. Local Subject–PCS correlations were less variable in Human 5+ than in Baseline at window sizes 5 and 10, but with only three contributors, no resampling method tested produced reliable confidence intervals for this ratio. The present pilot also showed lower between-session variability of lag +2 cross-correlations in the full Human condition, surviving correction within the 14-comparison lag family. However, the separate dataset did not reproduce the lag +2 pattern, and no lag survived its corresponding family correction.

These results should not be treated as mutually confirming evidence of a common phenomenon. The local-window and lagged analyses involved different Human samples, temporal scales, and estimands, and both were identified retrospectively. Their causes remain unknown, and findings from diagnostics available only in the separate dataset cannot be transferred automatically to the pilot. They instead demonstrate how the paired architecture can identify distinct forms of joint-stream structure and generate candidate estimands for prospectively specified analysis.

4.7 Architecture Parameters

The design choices reported here reflect one configuration among several the architecture supports. Three parameter classes are worth distinguishing because they are design settings rather than fixed features of the Paired-Delta Protocol.

Timing and partitioning. The timing of stream-label assignment relative to bit generation affects the mechanistic interpretation of a stream-specific difference. In

the present implementation, labels were determined from an assignment bit generated in the same call and applied after generation. A pre-assignment alternative would give the streams defined identities before their bits are generated. The present study does not compare these timing configurations, so the assignment-timing question discussed in Section 5.9 remains unresolved and should be addressed in a future prospectively designed comparison.

The method used to partition the generated sequence is a separate design choice. The present implementation divided each 300-bit sequence into two contiguous 150-bit halves, so the paired streams occupied different portions of the acquisition interval. Alternative configurations could distribute both streams across the full interval through interleaving or another time-balanced allocation. Such designs may reduce sensitivity to position-related changes within a call, but the retrospective diagnostic reported here does not establish that interleaving is generally superior. Stream partitioning should therefore be specified prospectively and evaluated under the intended acquisition conditions.

Measurement. The single-scale H_{RS} score was one choice of ordering metric, matched to the 150-bit block length used here (Section 4.5). The architecture does not privilege this metric, and alternatives may differ in their sampling behavior, sensitivity, precision, and bias. Stream length, blocks per session, calls per session, and audit frequency are also adjustable. The 150-bit length used here reflects feasibility constraints on session duration and API cost (Section 1.3) rather than a judgment that this length was optimal. The empirical physical-half

comparison in Section 3.2.2 applies specifically to the two contiguous 150-bit halves examined here and should not be generalized to other sequence lengths or partitioning rules.

Any stream-level metric can, in principle, be calculated separately for Subject and PCS and then examined through single-stream or paired-differential analyses.

Metrics defined jointly from the two streams can support relational analysis. Each metric and configuration requires separate calibration, including evaluation of finite-sample behavior, sensitivity, precision, and appropriate null distributions.

Task. Several features of task presentation can be changed without altering the paired architecture, including whether feedback is provided and how the target, feedback, or reward is presented. More substantial alternatives could change what information is available during acquisition, such as displaying the developing bit sequence in a ring or matrix rather than presenting only aggregate feedback. The task could also be framed around objectives other than intentional influence. These choices may affect engagement, learning, strategy, and the interpretation of any observed difference.

5. Limitations

5.1 Short protocol duration

Sessions were designed to contain 30 blocks, with two 150-bit streams generated per block. This brief format was constrained by available funding for participant compensation and QRNG API calls. It prioritized accessibility but limited statistical

resolution and increased the sampling variability of block-level metrics relative to protocols using longer streams or more blocks per session.

5.2 Post-hoc subgroup analysis

The Human 5+ subgroup was initially examined because participants recruited from the Reddit remote-viewing community had been encouraged to complete five to ten sessions, making session count an available but imperfect proxy for that recruitment group. The broader 2+ and 3+ thresholds were subsequently examined to explore whether the pattern might instead reflect repeated participation or learning. Human 5+ produced the strongest result among the three nested thresholds and is therefore emphasized here, with multiplicity across the threshold analyses addressed as described in Section 3.6.4.2. Because recruitment source, repeated participation, and self-selection are confounded, and Human 5+ includes only three participants, the subgroup finding remains exploratory.

5.3 Scope of the Exploratory-Search Calibration

The composition-conditional order-randomization analysis accounted for selection of the largest positive contrast across the three documented 2+, 3+, and 5+ session thresholds. The limitation is that this was a bounded calibration rather than an adjustment for the complete exploratory program. It did not include the wider historical search across metrics, populations, label-swap analyses, disjoint session-count groups, or relational measures. The calibration therefore strengthens interpretation of the result within the documented threshold family but

does not convert the Human 5+ finding into a confirmatory result. The relational analyses were also exploratory, with their search scope and multiplicity limitations described in Section 4.6.

5.4 Recruitment confound

Higher session-count participants were recruited from a different source than most single-session participants. Session count is therefore confounded with recruitment channel and participant characteristics. The present data cannot distinguish between exposure effects, trait differences, or self-selection.

5.5 Subgroup Size and Experimenter Participation

The 5+ session subgroup consists of three participants (~840 blocks), which limits what block-level analyses can establish about generalizability across observers. Clustered bootstrap resampling (Section 3.7.1) preserves the positive direction of the effect but widens confidence intervals substantially relative to block-level resampling, reflecting this small contributor count. One member of the subgroup was also involved in study administration. Although the paired architecture prevents mechanical influence on outcomes, inclusion of a non-blinded experimenter limits independence of the subgroup. Excluding this participant (Section 3.7.1) increases the effect size, because the principal investigator's blocks show the smallest individual paired-delta among the three.

5.6 Statistical resolution limits

As reported in Section 3.5, the Human-all versus Baseline comparison had approximately 80% power to detect a paired hit-rate contrast of 0.372 percentage points. This is substantially larger than the absolute hit-rate deviations of approximately 0.02 percentage points reported in the classical RNG-intention literature (e.g., Jahn et al., 1997; Bösch et al., 2006). This comparison provides only approximate scale context because the historical absolute deviation from chance and the present Human-minus-Baseline paired-delta contrast are different estimands.

The pilot was therefore not powered to reliably detect effects on that historical hit-rate scale. Its unresolved population-level results should not be interpreted as evidence that smaller effects are absent. More generally, underpowered individual studies have a low probability of detecting small effects even when such effects are present (Maxwell, Lau, & Howard, 2015).

Statistical resolution was also limited for the paired H_{RS} -score analysis. Under the session-cluster bootstrap location-shift model, the Human-all versus Baseline comparison had approximately 80% power to detect a contrast of 0.00405 H_{RS} -score units. The observed contrast was smaller (+0.00287), and its interpretation was borderline and analysis-method-sensitive: the hierarchy-preserving bootstrap interval only narrowly included zero, whereas the block-level Welch test did not

reach the conventional significance threshold (Section 3.6.1). The study therefore had limited power to reliably detect H_{RS} -score contrasts of the observed magnitude.

Greater precision requires larger samples, pooled evidence, or repeated independent studies, motivating the higher-powered preregistered replication described in Section 4.4.

5.7 AI control implementation

The AI condition combined GPT-4o-mini responses with scripted browser automation. Although completion of the model’s response gated each QRNG call, the agent could not deliberately pause, select an initiation time, or otherwise exercise the open-ended self-pacing available to human participants. It also had no persistent memory or model updating across blocks or sessions. The condition should therefore be interpreted as a constrained, API-driven procedural control rather than a fully adaptive agent with independent control over task timing or strategy.

5.8 QRNG Provider Heterogeneity and Execution Environment

QRNG calls were served through more than one API provider during the study period, but provider identity was not successfully recorded at the block level. Provider exposure therefore could not be balanced or included as an analytic factor. Paired subtraction removes provider-related contributions only when they are expressed equally in the two stream statistics; unequal, position-dependent, or condition-associated provider differences may remain. The three-subtest audit

battery showed similar pass rates across conditions, but its limited scope and the partial confounding of provider, date, and collection regime prevent it from excluding all provider-level differences.

Collection structure presents a related limitation. Baseline data were collected in larger batches, whereas Human data were generated through many short sessions occurring at different times. Consequently, condition was not fully separable from collection cadence and timing. The available audit and session-structure analyses found no corresponding instability or explanation for the observed subgroup pattern, but they did not directly test every possible effect of batch size, time of day, or longer-term drift.

Human participants also accessed the experiment through web browsers on uncontrolled local devices and networks. Latency and device differences were not explicitly modeled. Because both streams came from the same API response, paired subtraction would remove any resulting contribution expressed equally in both stream statistics, but it cannot exclude unequal or interaction-dependent effects.

Future studies should record provider identity for every block, balance provider exposure across conditions, and collect Baseline data in session-length batches staggered to match the timing and cadence of Human sessions. Device, browser, latency, and relevant timing variables should also be logged when privacy and feasibility permit.

5.9 Timing of stream assignment

The Subject and PCS labels were determined after generation of each 301-bit call. Bit 0 was generated in the same call as the two 150-bit halves and was subsequently read by the software to determine which physical half would be labeled Subject. Consequently, the halves had fixed physical positions during generation but did not yet have fixed Subject and PCS identities. Random assignment protects against a stable difference between the first and second halves systematically aligning with the Subject label, although it does not exclude chance imbalance, implementation error, or an artifact associated with the assignment bit itself.

This timing constrains the mechanistic interpretation of a stream-specific result. An observed difference aligned with the subsequently assigned Subject stream would require either an influence associated jointly with the assignment bit and the corresponding half, or some other process capable of coordinating the generated values with their later labels. By contrast, an influence acting uniformly on the complete call would not require stream identity to exist during generation, but it would ordinarily affect both subsequently labeled streams and cancel from the paired difference. Such a whole-call influence must therefore be investigated through direct-stream or reconstructed whole-call analyses rather than the paired delta alone.

A holistic model in which the assignment bit and both physical halves emerge as components of a jointly constrained outcome offers one possible interpretation, but

the present design cannot test that model or distinguish it from position-dependent, assignment-related, implementation-based, or chance explanations.

Assignment timing does not invalidate the label-swap sensitivity analysis reported in Supplementary Section S4. That test evaluates whether the observed alignment between the randomized labels and the paired stream scores is stronger than expected under label exchange. However, any genuine label-aligned result, whether identified by the original paired comparison or its label-swap equivalent, would inherit the same mechanistic timing question. Symmetric relational measures, such as same-window correlation or mutual information, do not depend on which stream is called Subject, whereas directional lagged relationships must be interpreted with care because exchanging the streams reverses the sign of the lag.

A design assigning the stream identities before generation would remove this particular ambiguity because the physical halves would already have Subject and PCS roles when their bits were produced. A direct comparison of pre-assigned and post-assigned conditions could therefore test whether assignment timing changes any observed stream-specific result.

5.10 Finite-Sample Behavior of the Single-Scale H_{RS} Score

The single-scale H_{RS} score is a transformed, finite-sample statistic derived from rescaled-range analysis and is used here as an ordering-sensitive measure. It is not a conventional multi-scale estimate of the Hurst exponent and should not be

interpreted as a precise or unbiased estimate of long-range dependence. Its behavior can depend on sequence length, bit composition, and the process producing the ordering structure. Consequently, equal estimator behavior under Baseline data does not establish that bias would remain equal if Subject and PCS had different underlying ordering structures.

A bounded simulation evaluated the exact 150-bit score using rank-binarized Gaussian AR(1) and fractional-Gaussian-noise sequences, while reproducing the observed Human 5+ Subject and PCS composition margins. The measured H_{RS} -score difference followed the imposed ordering contrast. It was positive when the stronger structure was assigned to Subject and negative when that structure was assigned to PCS. Across the examined parameter pairs, no resolved forward-versus-reverse distortion was detected. All pointwise 95% intervals for the directional asymmetry included zero, and the largest interval endpoint was approximately 0.00152, compared with the observed Human 5+ paired difference of +0.00435. Within these simulated cases, differential estimator asymmetry therefore did not account for the full observed contrast.

This calibration is nevertheless limited to the two generating models, parameter values, 150-bit sequence length, and composition margins examined. The process underlying the QRNG sequences is unknown, so the simulation neither supplies a bias correction nor proves that differential estimator behavior is absent under other forms of ordering structure. The observed paired H_{RS} -score difference should therefore be interpreted as an uncorrected, metric-specific contrast rather than as

a bias-free estimate of a latent difference in long-range dependence. This limitation concerns the selected metric rather than the paired-delta architecture, which can accommodate other calibrated stream-level or relational statistics. A related precision limit applies to the artifact-injection tests themselves: the persistence-mechanism injection in Section 3.3 could not rule out a residual reaching roughly half the observed Human 5+ contrast, for the same underlying reason, a limited number of realizations available at this pilot's scale. Mitigation for Exp 5 Pre-registration is discussed in Section 4.4.

5.11 Contiguous Stream Partitioning

The implemented protocol divided each 300-bit experimental sequence into two contiguous 150-bit halves, so the paired streams occupied different portions of the acquisition interval. Any position-related change within a call could therefore contribute unequally to the two stream statistics and would not necessarily be removed by their paired difference.

In a retrospective Baseline diagnostic, reconstructing the streams through locally balanced interleaving substantially attenuated the early-versus-late pattern observed under the contiguous split. The result was similar across 500 pseudorandom reconstructions and under a deterministic odd-even allocation. However, the alternative partition was examined after the pattern had been identified and used the same data in which it was first observed. The diagnostic therefore does not establish that the original pattern resulted from contiguous partitioning or that interleaving is generally superior. The absence of the same

resolved early-versus-late pattern in the Human and AI conditions also remains unexplained. Stream partitioning should be treated as a limitation of the pilot and evaluated prospectively using interleaved or otherwise time-balanced allocations. Interleaving would also more closely match the differential-measurement analogy introduced in Section 4.1. A conventional beam splitter divides one beam into two paths that traverse the same interval simultaneously, whereas the pilot's contiguous partition assigns Subject and PCS to sequential rather than simultaneous portions of the acquisition.

6. Conclusion and Recommendations

This study introduces a paired-delta QRNG architecture that retains conventional single-stream analysis while adding a same-call reference and relational analysis. Controlled injections demonstrated that specified equal-strength disturbances produced little systematic displacement of the paired difference, whereas stream-specific and unequal disturbances remained detectable. These tests confirmed the implemented operating behavior for the mechanisms examined but did not establish removal of a naturally occurring artifact. The empirical audits and Baseline diagnostics likewise supported the architecture under the sampled conditions without establishing universal protection against unequal, position-dependent, time-varying, nonlinear, or unmodeled contributions.

The paired hit-rate analyses detected no clear mean-level differences between conditions at the resolution of this pilot. The study had limited power to detect differences on the scale typically reported in the RNG-intention literature. The

unresolved results therefore do not establish equivalence or exclude smaller effects (Section 5.6).

Separately, an exploratory subgroup of three participants who completed at least five sessions showed small nominal distributional differences from Baseline in both the direct Subject-stream H_{RS} -score analysis and the paired Subject-minus-PCS H_{RS} -score analysis. These analyses used the same underlying stream scores and are not independent confirmation. The 5+ result was the strongest among the retrospectively examined 2+, 3+, and 5+ session thresholds. A composition-conditional simulation that repeated this three-threshold selection process produced a joint-max tail area of .008, indicating that a maximum contrast this large was uncommon under the simulated null. However, this calibration addressed only selection among these three thresholds and did not account for the wider exploratory search across other populations, metrics, and analyses. Moreover, its bidirectional configuration is compatible both with differential stream behavior and with retrospective selection for a large paired difference (Section 3.6.4.2). It is therefore presented as an exploratory worked example rather than an established effect and requires preregistered replication.

The primary contribution is increased diagnostic leverage. Same-call paired streams retain the conventional Subject-stream comparison while revealing how Subject, PCS, and the empirical Baseline contribute to an observed contrast. They also make relational properties measurable, including the association, temporal alignment, stability, and joint distribution of the two streams. These analysis

families are not independent confirmations, but together they can prevent ambiguous findings from being mistaken for stronger evidence than they provide and can reveal patterns that a single-channel design cannot observe. The exploratory relational analyses illustrate this capability and identify candidate estimands for prospective testing without establishing a condition-specific alteration of Subject–PCS coupling. Ordering-sensitive metrics further extend the architecture beyond conventional mean hit-rate analysis, although each metric requires configuration-specific calibration and interpretation.

Progress in this area requires not only stronger controls but experimental structures that make otherwise inaccessible distinctions and relationships measurable. By generating a paired companion stream from the same physical draw, PDP provides conditional control over shared contributions, reveals how a contrast is assembled, and makes the relationship between streams observable. This relatively simple modification both expands the questions that can be asked of each draw and improves scrutiny of conventional findings. Combined with preregistration, adequate sample sizes, fixed participation schedules, and independent replication, it provides a framework for more discriminating investigations of subtle and uncertain phenomena.

Acknowledgements and Disclosures

In memory of my father, Alfred C. Rester Jr. (1940–1996), nuclear physicist and arms-control scientist, whose work ranged from the Gamma Ray Advanced

Detector to the observation of Supernova 1987A, and whose love of discovery inspired my own.

AI-Assisted Research

Large language models were used during the development of experimental code, data processing pipelines, and manuscript drafting. The study design, architectural decisions, and analytic strategy were directed by the author. All generated code and analyses were reviewed and validated against the frozen dataset prior to inclusion in the manuscript.

Data Transparency

All data collection procedures and metrics were defined in the experiment source code prior to analysis. The dataset was frozen using SHA-256 hashing before statistical evaluation. Raw data and full computational notebooks are available in the public repository referenced in the Data Availability section.

Ethics and Informed Consent

The study was conducted by an independent researcher without institutional affiliation, and prospective review by an institutional review board or research ethics committee was not obtained. The study was conducted with reference to the principles of respect for persons, beneficence, and justice described in the Belmont Report (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979). Participation was voluntary and occurred through

an online experimental interface. Participants were informed about the study before choosing to participate. The noninvasive computer-based task typically required approximately 3–5 minutes, and participants could leave the study at any time. No names, contact information, IP addresses, or other direct personal identifiers were collected through the experimental interface. Participants were provided with a study-generated session identifier that could be used to request their results or removal of the corresponding session data. Given the brief, noninvasive procedure, voluntary participation, and absence of direct identifying information from the experimental dataset, the study was considered to present limited foreseeable risk. This assessment was made by the author and was not a formal determination by an independent ethics committee.

Conflict of Interest

The author declares no financial interest in any QRNG hardware, software platform, or related commercial product. This work was conducted independently and without external funding.

Data Availability Statement

The raw datasets, the frozen data manifest (SHA-256), and the Python analysis notebooks used to generate the findings in this study are available in the public repository at [10.5281/zenodo.20763628](https://doi.org/10.5281/zenodo.20763628). The experimental source code (JavaScript), including the pre-specified [hurstApprox](#) and delta logic, is also provided on [GitHub](#) to ensure full transparency and replicability.

Experiment 5 is preregistered at

https://osf.io/yseqj/overview?view_only=98fef7fec86244749d80e778424a065

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Supplement

This Supplement retains detail cut from the main text during compression of Sections 3.6 and 3.7. Nothing here changes the status of any result in the main text. Items retained here are exploratory, non-confirmatory, and in most cases already summarized with their conclusion stated in the main text. This Supplement does not introduce new claims.

S1. Full Position, Weighting, Recurrence, and Session-Order Detail (expands Section 3.7.1)

This section asks whether the Human 5+ pattern is stable across within-session position, contributor/session weighting scheme, individual sessions, and time (first vs. later sessions) and whether it depends on any one of these.

S1.1 Within-Session Position

Within-session position was first examined by estimating the change in the paired Subject-minus-PCS single-scale H_{RS} score from the beginning to the end of each complete session. The estimated slopes were +0.00553 for Baseline (95% descriptive interval [$-0.00244, +0.01350$]), -0.00178 for AI ($[-0.01235, +0.00929]$), -0.00137 for all Human ($[-0.00843, +0.00533]$), and

+0.00008 for Human 5+ ($[-0.01212, +0.01250]$). All intervals included zero and no group showed a resolved within-session trend.

A second, post hoc check asked whether the Human 5+ separation was concentrated in a particular part of the session rather than distributed evenly. After the negative realized Baseline mean was identified as contributing to the Human 5+–Baseline contrast, a post-hoc diagnostic divided each retained session into early blocks 0 through 14 and late blocks 15 through 29. Baseline's early mean was -0.004065 , compared with $+0.000583$ in its late blocks. The early-minus-late Baseline contrast was -0.004647 , with a session-clustered 95% interval excluding zero at the unadjusted level. The corresponding contrasts were unresolved in Human and AI, including when AI uncertainty was clustered across its 15 independent collection bursts. None of the three condition-specific contrasts remained resolved under the Bonferroni threshold of $0.05/3=.0167$.

Within Human 5+, the early mean was $+0.005857$ and the late mean was $+0.002842$. The Human 5+–Baseline contrast was $+0.009921$, 95% CI $[+0.002908, +0.021763]$, in the early half and $+0.002259$, 95% CI $[-0.005471, +0.014116]$, in the matched late half. The late-minus-early change in the condition contrast was -0.007662 , with a 95% interval of $[-0.019489, +0.004125]$. Thus, the observed separation was concentrated in the early half, but the direct interaction did not resolve a difference between halves. This split-half analysis was post hoc and does not establish a systematic position effect.

S1.2 Session Structure and Weighting Sensitivity

To examine sensitivity to contributor weighting, the Human 5+ versus Baseline paired H_{RS} score contrast was summarized under two distinct weighting methods.

Block weighting produced a contrast of +0.00609. Equal-session weighting produced the identical value because every retained session contributed 30 blocks. It therefore did not provide a separate sensitivity result in this dataset.

Giving each of the three Human 5+ contributors equal weight, and then weighting their sessions equally, produced a contrast of +0.00818. The contrast therefore held direction and did not collapse toward zero under any weighting scheme tested, though the three schemes are not independent of one another given the same three contributors.

A separate hierarchy-preserving bootstrap used 10,000 draws. Human resampling selected contributors and then complete sessions within each selected contributor; Baseline resampling selected complete sessions. Blocks within a session were always retained together. For the block-weighted contrast, the resulting descriptive 95% interval was [+ 0.00094, + 0.01585], excluding zero under this resampling scheme.

S1.3 Session Recurrence and PI-Influence Sensitivity

The Human 5+ subgroup contained three contributors and 28 complete sessions. Fifteen sessions had positive paired Subject-minus-PCS H_{RS} scores and 13 had negative scores, indicating the positive mean was not driven by a small number of

extreme sessions. The median session-level paired score was +0.003492, with an interquartile range from -0.007743 to +0.015702.

Leaving out each complete session in turn produced pooled paired scores ranging from +0.003270 to +0.005046, compared with +0.004349 in the full subgroup. No single session was driving the result.

The principal investigator contributed 18 of the 28 sessions. Excluding the principal investigator increased, rather than decreased, the block-weighted paired score (+0.004349 to +0.008682) and the Human 5+–Baseline contrast (+0.006090 to +0.010423), indicating the pattern was not an artifact of the non-blinded contributor's involvement.

S1.4 First Versus Later Sessions

A descriptive within-contributor analysis compared each eventual Human 5+ contributor's first complete session with their mean across later sessions. The equal-contributor mean later-minus-first change in paired H_{RS} score was +0.004560 but this average obscured divergent individual patterns. Two contributors increased (range -0.013107 to +0.014535) and one decreased. With only three contributors, this analysis cannot distinguish a genuine within-person trend from individual variability.

S2. Full Local and Lagged Coupling Detail (expands Section 3.7.2)

This section asks whether Subject and PCS streams show reduced local variability or lagged coupling beyond what's expected by chance, and whether any such pattern replicates in a separate dataset.

S2.1 Local Subject–PCS Coupling Stability

Local Subject–PCS coupling was calculated within nonoverlapping windows of five and ten blocks from each complete session. Larger window sizes were ignored for this dataset as 30-block sessions are insufficient for a within-session variance estimate. Neither the mean window-level covariance nor the mean window-level correlation clearly differed between Human 5+ and Baseline. Window-level correlations varied less in Human 5+ than in Baseline: the variance ratios were 0.832 at $W = 5$ and 0.742 at $W = 10$, corresponding to approximately 17% and 26% lower observed variance, a candidate pattern rather than a resolved one. With only three independent contributors in this subgroup, confidence intervals for this ratio are not reliable under any resampling method tested, cluster-based or unclustered. The analogous covariance-variance point estimates were also below 1. This analysis was retrospective.

A qualitatively similar point-estimate pattern was observed in a separately collected test dataset with 80-block sessions, where the larger session length permitted testing at $W = 15$ and $W = 20$ as well. In that dataset, the pattern at these longer window sizes was present in the full Human population and was not confined to

any subgroup. A parallel comparison of the full Human population against Baseline in the primary pilot, limited to $W = 5$ and $W = 10$ by the fixed 30-block session length, showed no resolved difference. The primary pilot's 30-block sessions cannot support window sizes beyond $W = 10$, so it cannot confirm or rule out whether this pattern strengthens at longer windows. The separate dataset was not used for confirmatory inference here, and whether the longer-window pattern reflects a real effect, a property of that dataset, or chance remains an open question for the preregistered replication.

S2.2 Lagged Subject–PCS Coupling

Session-level cross-correlations between Subject and PCS H_{RS} scores were examined at lags from -3 through $+3$. A positive lag indicates that Subject at block t was correlated with PCS at block $t + k$. The analysis compared Human and Baseline sessions using seven mean comparisons and seven variance comparisons, forming a 14-comparison family. Only the variance comparison at lag+2 survived the Bonferroni threshold of $0.05/14 = .00357$. At this lag, Human correlations had approximately 45% lower observed variance than Baseline (*variance ratio* = 0.5505, Levene $p = 0.00110$; hierarchy-preserving sensitivity interval: 0.383–0.808). A sensitivity analysis resampling Human contributors and their complete sessions hierarchically, and Baseline sessions as complete units, found the lag+2 result remained unusual both individually ($p = 0.00170$) and after maximum-statistic adjustment across the seven variance comparisons ($p = 0.00780$). A further conservative adjustment covering all 14 mean and

variance comparisons gave $p = 0.01560$, so the lag+2 variance result held up under the strictest correction applied. The corresponding mean-correlation difference was unresolved ($+ 0.0354$, $p = 0.154$), and no other comparison survived family adjustment.

This correction covers only the 14-comparison lag family and does not encompass the broader collection of exploratory analyses conducted in the pilot, so this result should not be read as confirmatory. No mechanism specifically predicted a two-block displacement, and differences in participant composition, collection cadence, score distributions, or temporal structure could affect the variability of session-level correlations.

S3. Shuffle Sensitivity of the H_{RS} Score

This section asks whether the Human 5+ vs. Baseline separation depends on within-block bit ordering, as opposed to bit composition alone.

The H_{RS} score is calculated from the cumulative-deviation path of a sequence.

Reordering the same bits can therefore change the score even when bit composition is unchanged.

For each paired block, the same positional permutation was applied to the Subject and PCS streams within non-overlapping windows. This preserved the bit composition, pairing, and shared positional transformation while changing within-stream ordering. Window sizes were 1, 3, 5, 10, 25, 50, and 150 bits. At $W =$

1, the ordering was unchanged. At $W = 150$, each complete 150-bit stream was randomly reordered. The procedure was repeated 200 times for each nontrivial window size.

The unshuffled KS distance between the Human 5+ and Baseline paired H_{RS} -score distributions was 0.0541. For each nontrivial window size, the table reports the mean KS distance across 200 shuffle realizations.

Table S1. Mean Human 5+ versus Baseline KS distances after composition-preserving paired-stream shuffling

Window size	Mean KS distance
1, unshuffled	0.0541
3	0.0491
5	0.0600
10	0.0628
25	0.0586
50	0.0395
150, full-block shuffle	0.0333

The response was not monotonic across window sizes, indicating the score's ordering-sensitivity does not scale simply with window size. The smallest mean distance occurred after full-block shuffling. These results show that the measured distributional separation changed when within-block ordering was altered while bit composition was held constant. They demonstrate that the H_{RS} score responds to ordering at the implemented 150-bit length.

This is a sensitivity analysis of the score, characterizing how the H_{RS} score responds to ordering; the Human 5+ comparison itself is evaluated in Section 3.6.4.

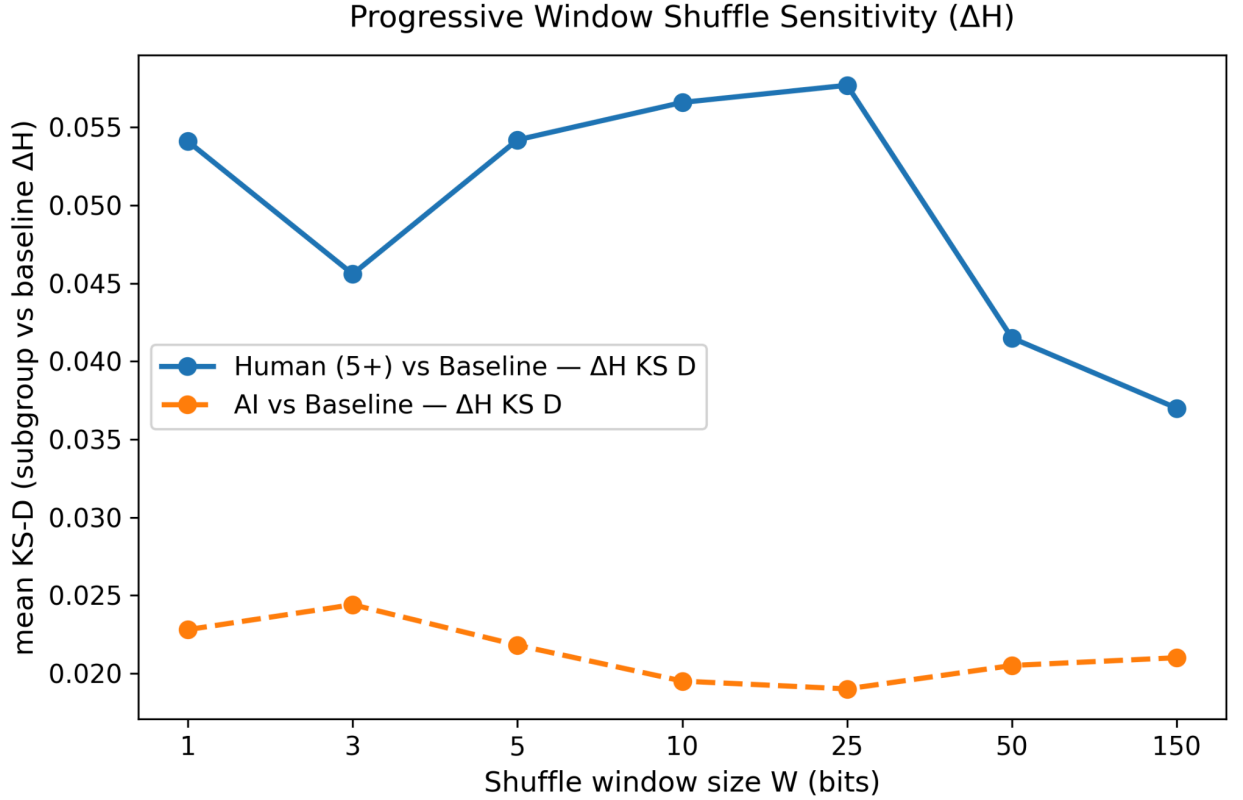


Figure S1. Sensitivity of the Human 5+ versus Baseline distributional separation to within-block shuffling.

The figure shows the mean KS distance between the paired H_{RS} score distributions after the same within-window positional permutation was applied to both streams of each block. Each nontrivial window size was evaluated using 200 realizations. $W = 1$ is the unshuffled reference and $W = 150$ is

a full-block shuffle. The figure is descriptive and is not a participant-level hypothesis test.

S4. Session-Count Thresholds and Stream-Level Decomposition

This section asks whether the Human 5+ result is an artifact of which session-count threshold was chosen, and whether it arises more from the Subject stream, the PCS stream, or their difference.

The Human 5+ threshold was initially examined because participants recruited through the Reddit remote-viewing community had been encouraged to complete five to ten sessions. The 2+, 3+, and 5+ thresholds were subsequently examined as retrospective comparisons.

These thresholds were nested. Participants in the 5+ group were also included in the 2+ and 3+ groups. The comparisons therefore did not represent independent participant groups. The disjoint 2–4 group was examined separately as a follow-up comparison.

Table S2. Session-count threshold comparison

Human session count group	Contributors	Sessions	Blocks	Mean paired H_{RS}	Unadjusted KS p
2+ sessions	10	47	1,410	+0.00389	.0577
3+ sessions	6	39	1,170	+0.00341	.0496

5+ sessions	3	28	840	+0.00435	.0398
2–4 sessions, disjoint follow-up	7	19	570	+0.00322	.244

Table S2. The 2+, 3+, and 5+ groups are nested, overlapping thresholds rather than independent groups. Their block-level KS values are retained as unadjusted descriptive results. The disjoint 2–4 group excludes the three 5+ participants and was examined separately as a follow-up comparison. The composition-conditional order-randomization analysis reported in Section 3.6.4.2 provides the documented threshold-search calibration.

The 5+ comparison was the strongest of the nested thresholds, but because the same three 5+ participants contributed to all three nested groups, the pattern across the 2+, 3+, and 5+ thresholds should not be interpreted as an independent dose-response gradient. The disjoint 2–4 group, which excludes those three participants entirely, did not show a clear distributional difference from Baseline ($p = .244$) and the pattern did not extend to Human participants outside the specific subgroup that generated it.

Separate Subject and PCS comparisons were also examined to describe how the paired contrasts arose. The paired t-test and label-swap test in Table S3 evaluate the same mean paired HRS contrast against Baseline. They are alternative analyses of the same underlying contrast rather than independent evidence.

Table S3. Stream-level directional pattern by cohort

Cohort	p_t	p_{swap}	Subject vs Baseline	PCS vs Baseline
Human, 1 session	0.287	0.287	↑ ns	↔ ns
Human, 2–4 sessions	0.090	0.085	↑ marginal	↔ ns
Human, 5+ sessions	0.0147	0.014	↑ marginal	↓ marginal
AI Agent	0.084	0.086	↔ ns	↔ ns

Table S3. The paired t-test and label-swap test evaluate the same mean paired HRS contrast against Baseline. The label-swap test randomly exchanges Subject and PCS labels within blocks. The Subject and PCS columns show the direction of each stream relative to Baseline based on separate stream-level tests: ↑ indicates higher, ↓ indicates lower, and ↔ indicates no consistent direction. “Marginal” indicates $0.05 < p < 0.10$, $ns = p \geq 0.10$.

The tests in Table S3 treat blocks as independent and are reported as exploratory descriptions with unadjusted p-values. The opposed Subject and PCS directions in Human 5+ describe the arithmetic components of the paired contrast, not separate confirming evidence or a stream-specific mechanism.

References

Anis, A. A., & Lloyd, E. H. (1976). The Expected Value of the Adjusted Rescaled Hurst Range of Independent Normal Summands. *Biometrika*, 63(1), 111–116.

<https://doi.org/10.2307/2335090>

Bösch, H., Steinkamp, F., & Boller, E. (2006). Examining psychokinesis: The interaction of human intention with random number generators--A meta-analysis. *Psychological Bulletin*, 132(4), 497–523.

<https://doi.org/10.1037/0033-2909.132.4.497>

Grote, H. (2017). Multiple-Analysis Correlation Study between Human Psychological Variables and Binary Random Events. *Journal of Scientific Exploration*, 31(2), 231–254.

Hamed, K. H. (2007). Improved finite-sample Hurst exponent estimates using rescaled range analysis. *Water Resources Research*, 43(4), W04413.

<https://doi.org/10.1029/2006WR005111>

Jahn, R. G., Dunne, B. J., Nelson, R. D., Dobyns, Y. H., & Bradish, G. J. (1997). Correlations of random binary sequences with pre-stated operator intention: A review of a 12-year program. *Journal of Scientific Exploration*, 11(3), 345–367.

Maier, M. A., Dechamps, M. C., & Pflitsch, M. (2018). Intentional observer effects on quantum randomness: A Bayesian analysis reveals evidence against

micro-psychokinesis. *Frontiers in Psychology*, 9, 379.

<https://doi.org/10.3389/fpsyg.2018.00379>

Maxwell, S. E., Lau, M. Y., & Howard, G. S. (2015). Is psychology suffering from a replication crisis? What does “failure to replicate” really mean? *American Psychologist*, 70(6), 487–498. <https://doi.org/10.1037/a0039400>

Nelson, R.D. & Bancel, P.A. (2011). "Effects of mass consciousness: Changes in random data during global events." *Explore: The Journal of Science and Healing*, 7(6), 373–383.

Radin, D., Nelson, R. (1989). Evidence for Consciousness-Related Anomalies in Random Physical Systems. *Foundations of Physics* 19(12), 1499-1514.

Radin, D., Nelson, R., Dobyns, Y., & Houtkooper, J. (2006). Reexamining psychokinesis: Comment on Bösch, Steinkamp, and Boller (2006). *Psychological Bulletin*, 132(4), 529–532. <https://doi.org/10.1037/0033-2909.132.4.529>

Walach, H., Kirmse, K. A., Sedlmeier, P., Vogt, H., Hinterberger, T., & von Lucadou, W. (2021). Nailing Jelly: The Replication Problem Seems to Be Unsurmountable. *Journal of Scientific Exploration*, 35(4), 788–828.

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on *p*-values: Context, process, and purpose. *The American Statistician*, 70(2), 129–133. <https://doi.org/10.1080/00031305.2016.1154108>

Wilson, D. B., & Shadish, W. R. (2006). On blowing trumpets to the tulips: To prove or not to prove the null hypothesis. Comment on Bösch, Steinkamp, and Boller (2006). *Psychological Bulletin*, 132(4), 524–528.

<https://doi.org/10.1037/0033-2909.132.4.524>